Representative Learning for Distributed Support Vector Machines

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Abstract

Distributed support vector machine (SVM) learning is challenging under heterogeneous local data and limited communication, particularly because the nonsmooth hinge loss complicates gradient-based optimization and the construction of compact summaries of local objectives. We develop a Representative Learning framework based on Lifted Empirical Risk Minimization, in which each covariate is replaced by a local Gaussian distribution. For linear SVMs, the resulting expected hinge loss has an exact closed form that is smooth for nondegenerate lifting covariances and converges to the ordinary hinge loss as the covariance vanishes.

The lifted formulation leads naturally to clouded representatives that encode both location and covariance information. We study a clouded Mean Representative (MR), which summarizes class-wise local data, and a clouded Score-Matching Representative (SMR), which additionally matches the class-wise gradient of the lifted objective at its construction parameter. For SMR, we show that the representative covariance scale remains close to the original scale as within-group covariate heterogeneity decreases, and that the discrepancy between the full-data gradient and the gradient evaluated on frozen representatives is controlled by the largest within-group diameter. Under strong convexity and smoothness, this yields an inexact-gradient convergence guarantee whose error floor is proportional to the representative approximation error.

Synthetic experiments under heterogeneous class allocation show that the lifted representative methods remain competitive with centralized solutions when parameter-aggregation methods are affected by repeated local optimization. Although MR can achieve similar predictive accuracy, SMR more closely recovers the centralized lifted-SVM optimizer. We further extend the framework to nonlinear SVMs through a shared kernel dictionary that provides a common finite-dimensional coordinate system for constructing and communicating representatives. Experiments on the Ionosphere and Australian Credit Approval datasets show analogous behavior in the kernel setting. Together, these results provide a distributional representative framework for communication-efficient distributed large-margin learning under heterogeneity.

Keywords: support vector machines; distributed learning; representative learning; Lifted Empirical Risk Minimization; clouded representatives; data heterogeneity

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1 Introduction

Distributed Machine Learning (DML) enables learning from data distributed across multiple nodes without requiring the raw data to be centralized. Federated Learning (FL) provides a prominent framework for this setting, in which models are trained locally and information is communicated across nodes or through a coordinating server (McMahan et al., 2017; Kairouz and McMahan, 2021). Despite substantial progress, distributed learning remains challenging when data are heterogeneous and communication is constrained. Local datasets are often non-IID, with potentially substantial differences in feature and response distributions, so locally trained models or updates may provide conflicting information about the global learning objective. At the same time, privacy and data-localization requirements can prohibit direct exchange of raw observations, while limited bandwidth, intermittent connectivity, and asynchronous participation restrict the frequency and volume of communication. Consequently, a fundamental question in DML is not only how to optimize a learning objective across distributed data, but also what information should be communicated between nodes to support accurate and robust learning under these constraints.

Support Vector Machines (SVMs) provide an important setting for studying distributed classification. As classical margin-based classifiers, SVMs offer strong predictive performance, a convex training objective, and flexible nonlinear decision boundaries through kernel methods (Cortes and Vapnik, 1995; Vapnik, 1999). Their relevance also extends beyond classical kernel learning. Modern neural networks with rectified linear unit (ReLU) activations construct complex piecewise-linear prediction functions, creating useful conceptual connections between margin-based learning and modern deep architectures. Thus, SVMs provide a mathematically tractable setting in which the effects of data geometry, margins, and distributed information sharing can be studied explicitly, while retaining connections to broader classes of modern learning models.

At the same time, SVMs pose distinctive challenges in distributed environments. Their empirical objective is based on the nonsmooth hinge loss, and the learned decision boundary depends critically on observations near or within the margin. When data are distributed across heterogeneous nodes, the observations that are locally important for defining a decision boundary may differ substantially across nodes. One natural strategy is therefore to communicate support vectors, which characterize the SVM solution, but support vectors are themselves observations from the local datasets and may constitute a substantial portion of the data Graf et al. (2004). Other distributed SVM approaches coordinate local classifiers through repeated exchange of model parameters or consensus updates, for example through ADMM-based formulations (Forero et al., 2010; Zhang and Zhu, 2018). Kernel SVMs introduce an additional difficulty because nonlinear decision functions may be represented through data-dependent kernel expansions, motivating approximations such as Nyström methods or Random Fourier Features when explicit finite-dimensional representations are needed Williams and Seeger (2000); Rahimi and Recht (2007). These characteristics make distributed SVM learning a useful testbed for asking how the information needed to determine a global decision boundary can be communicated compactly without directly transmitting the underlying observations or relying on frequent model exchange.

Representative Learning (RepL) provides an alternative approach to distributed communication by replacing raw observations or repeated model updates with compact pseudo-data representatives constructed from local datasets (Li and Yang, 2022; Zheng et al., 2024; Li, 2026). Existing RepL methods have primarily focused on point-valued representatives, including mean- and score-based constructions that preserve selected statistical or optimization information from local data. While effective in many settings, point representatives may not adequately capture local variability and geometric information that can be particularly important for nonsmooth learning problems.

To address this limitation, Wang and Li (2026) introduced *clouded representatives*, extending RepL from point-valued to distribution-valued representatives, together with a Lifted Empirical Risk Minimization (Lifted ERM) framework in which individual observations are replaced by local probability measures and learning proceeds through expected loss over these distributions. Classical ERM is recovered as the special case in which the probability measures degenerate to point masses. Originally developed for distributed quantile regression, this framework provides a natural foundation for extending RepL to other nonsmooth learning problems. In this work, we build on the Lifted ERM and clouded-representative framework of Wang and Li (2026) to develop representative learning for distributed SVMs with hinge loss.

Our contributions are fourfold.

(i) We formulate a lifted empirical risk for SVMs in which observations are represented by local probability distributions and derive the resulting lifted hinge loss for Gaussian clouds.

(ii) We construct clouded representatives for distributed linear SVMs and characterize their approximation to the corresponding local empirical risks/gradients.

(iii) We extend the framework to nonlinear classification through RKHS representations and kernel approximations.

(iv) We evaluate the resulting distributed algorithms under heterogeneous node distributions using synthetic and benchmark datasets.

2 Background and Literature Review

DML considers learning from data partitioned across multiple computational nodes, with distributed optimization methods ranging from parallel stochastic gradient descent and parameter-server architectures to modern FL (Zinkevich et al., 2010; Li et al., 2014; Konečnỳ et al., 2016). FL emphasizes settings in which data are naturally distributed across devices or institutions and raw observations remain local, with Federated Averaging (FedAvg) providing a canonical approach based on alternating local optimization and global parameter aggregation (McMahan et al., 2017; Kairouz and McMahan, 2021). A central challenge is that local data are generally non-IID, so repeated local optimization can introduce client drift and degrade convergence or statistical accuracy (Zhou and Cong, 2018; Haddadpour and Mahdavi, 2019; Khaled et al., 2020; Koloskova et al.,

2020; Li et al., 2020). A substantial literature consequently modifies local or global optimization to improve robustness to heterogeneity, including proximal and variance-corrected methods, local-update and communication-efficient SGD, and robust aggregation or sampling strategies (Li et al., 2020; Karimireddy et al., 2020; Stich, 2018; Stich and Karimireddy, 2020; Spiridonoff et al., 2021; Wang and Joshi, 2021; Wang et al., 2021). Related approaches relax the requirement of a single global model through personalized and multi-task federated learning (Fallah et al., 2020; Li et al., 2021; Tan et al., 2022; Smith et al., 2017; Argyriou et al., 2006; Zhang and Yeung, 2012), while decentralized methods replace centralized aggregation by peer-to-peer communication (Lian et al., 2017; Assran et al., 2019). Another line reduces dependence on parameter exchange through knowledge distillation, communicating model predictions rather than complete model states and thereby accommodating, in some settings, heterogeneous local architectures (Hinton et al., 2015; Chang et al., 2019; Lin et al., 2020; Wu et al., 2022; Zhu et al., 2021). Distributionally robust optimization provides a complementary approach by optimizing against distributional perturbations to improve robustness under distribution shift (Kuhn et al., 2019). Despite these developments, most distributed learning methods communicate quantities derived from the current model, such as parameters, gradients, or predictions, and their effectiveness can remain sensitive to data heterogeneity, communication frequency, model compatibility, and synchronization. These limitations motivate communication mechanisms that summarize the local data itself rather than only the state or output of a local model.

Data heterogeneity is one of the central challenges in distributed learning because local datasets are rarely independent and identically distributed across nodes. Heterogeneity can arise through differences in feature distributions $P_k(X)$, label distributions $P_k(Y)$, conditional relationships $P_k(Y|X)$, or class-conditional feature distributions $P_k(X|Y)$, corresponding broadly to feature, label, model, and representation heterogeneity. These differences cause local objectives and local update directions to deviate from the global learning objective, so algorithms based on repeated local optimization and parameter aggregation can suffer from client drift, slower convergence, instability, and degraded global performance Zhou and Cong (2018); Haddadpour and Mahdavi (2019); Khaled et al. (2020); Koloskova et al. (2020); Li et al. (2020); Karimireddy et al. (2020). The problem becomes particularly pronounced when nodes have severe class imbalance, distribution shift, or substantially different local prediction relationships. A broad literature has therefore sought to mitigate heterogeneity through proximal or control-variate corrections, robust aggregation, personalized learning, multi-task learning, and distributionally robust optimization. More generally, heterogeneity changes the goal of distributed learning from simply reproducing centralized optimization to determining how information from statistically different nodes should be combined so that collaboration remains beneficial.

2.1 Representative Learning: From Point to Cloud

RepL provides a data-centric framework for distributed learning in which nodes communicate compact pseudo-data, called *representatives*, rather than repeatedly exchanging model parameters

or gradients Li and Yang (2022); Li (2026).

Suppose the full dataset is distributed across K nodes,

$$\mathcal{D} = \cup_{k=1}^{K} \mathcal{D}_k, \quad \mathcal{D}_k = \{(\mathbf{x}_i, y_i)_{i \in I_k}\},$$

where I_k is the index set for node k and $n_k = |I_k|$ is its sample size. A weighted point representative of $\mathcal{D}_k$ takes the form $\tilde{\mathcal{D}}_k = (n_k, \tilde{\mathbf{x}}_k, \tilde{y}_k)$, constructed from the local data.

More generally, a node may be divided into r_k subpartitions, giving the representative dataset

$$\tilde{\mathcal{D}}_k = \left\{(n_{kh}, \tilde{\mathbf{x}}_{kh}, \tilde{y}_{kh})_{h=1}^{r_k}\right\}$$

The collection $\tilde{\mathcal{D}} = \bigcup_{k=1}^{K} \tilde{\mathcal{D}}_k$ then provides a compact pseudo-dataset on which downstream learning can be performed. Different representative constructions differ primarily in which statistical or optimization information of the original data they seek to preserve.

The earliest RepL constructions are point-valued representatives. The *Mean Representative* (MR) provides a model-free summary by taking the coordinatewise means of the observations within a local group Li and Yang (2022). MR is particularly natural for linear models, but direct averaging does not generally preserve the nonlinear relationship between the response and the linear predictor in generalized linear models (GLMs). To better account for this nonlinear structure, the *Transformed Mean Representative* (TMR) constructs the response representative in the transformed space Li (2026). TMR therefore retains the simplicity of a mean-based representative while incorporating model-specific transformation information.

A different construction principle is provided by the *Score-Matching Representative* (SMR), which constructs representatives to match local score function values in a GLM Li and Yang (2022). Thus, unlike MR and TMR, SMR is explicitly model-oriented and encodes information from the local score function into the representative. The *Response-Aided Score-Matching Representative* (RASMR) Zheng et al. (2024) refines heterogeneous local blocks using response- and model-related information, to improve the construction of representative points. The *Anchored Score-Matching Representative* (Anchored-SMR) addresses the original underdetermined score-matching system by matching the intercept component of the score. Anchored-SMR provides an identifiable score-matching construction with a fixed intercept representation.

Despite these developments, the representative constructions above remain point-valued: a collection of observations is ultimately summarized by one or more weighted pseudo-observations. Compression to isolated points can discard information about within-group variability and local geometry, which may be particularly important for nonsmooth learning objectives.

To incorporate such information, Wang and Li (2026) introduced *Lifted Empirical Risk Minimization* (Lifted ERM) and *clouded representatives*, extending point-based RepL to distribution-valued observations. Let $\ell(f_\theta(\mathbf{x}), y)$ denote a loss function evaluated under a predictive model f_θ. Each observed covariate $\mathbf{x}_i$ is associated with a local probability measure Q_i, and the lifted empirical risk

is defined as

$$R_n^Q(\theta) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}_{\mathbf{X}_i \sim Q_i}\left[\ell\left(f_\theta(\mathbf{X}_i), y_i\right)\right]. \tag{1}$$

When $Q_i = \delta_{\mathbf{x}_i}$, with $\delta_{\mathbf{x}_i}$ denoting the Dirac measure at $\mathbf{x}_i$, equation (1) reduces to the classical empirical risk. Lifted ERM therefore embeds conventional point-based ERM as a special case while permitting learning over local probability distributions rather than isolated observations Wang and Li (2026).

Within this framework, Wang and Li (2026) introduced a *clouded representative* as a distributional extension of a point representative. Rather than representing a local group only through a response and covariate location, a clouded pseudo-sample additionally carries distributional information describing local variability. For distributed quantile regression, Wang and Li (2026) develops clouded representatives containing representative response, location, and covariance information, and constructs Clouded SMR by matching score equations under the lifted risk. This distribution-valued representation preserves local geometric and variability information that is absent from a point summary and is especially relevant for nonsmooth objectives, where expected loss can depend on how probability mass is distributed around the nondifferentiable region of the loss.

2.2 Distributed Support Vector Machines

For observations $(\mathbf{x}_i, y_i)$, with $\mathbf{x}_i \in \mathbb{R}^p$ and $y_i \in \{-1, 1\}$, a linear classifier takes the form $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$. The soft-margin SVM can be expressed through the regularized hinge-loss problem

$$\min_{\mathbf{w},b} \ \frac{\lambda}{2}\|\mathbf{w}\|_2^2 + \frac{1}{n}\sum_{i=1}^{n} \max\left\{0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)\right\}. \tag{2}$$

The hinge loss is convex but nonsmooth at observations satisfying $y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$. This nonsmooth structure distinguishes SVM optimization from smooth likelihood-based problems and commonly motivates subgradient, coordinate-descent, or quadratic-programming methods.

In distributed settings, if the objective is to recover a single global classifier, the centralized SVM objective can be decomposed into node-level hinge-loss contributions,

$$\min_{\mathbf{w},b} \ \frac{\lambda}{2}\|\mathbf{w}\|_2^2 + \frac{1}{n}\sum_{k=1}^{K}\sum_{i \in I_k} \max\left\{0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)\right\}. \tag{3}$$

The statistical target is therefore well defined even when the observations cannot be pooled. The central difficulty is how to solve (3) when each node has access only to its local data and communication between nodes is restricted.

Existing distributed SVM methods approach this problem through different choices of communicated information. One natural strategy is to exchange support vectors. From the dual representation, $\mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i$, only observations with $\alpha_i > 0$ contribute directly to the classifier.

These support vectors therefore provide a natural data-based summary of local SVM solutions and have been used in hierarchical distributed training schemes. However, support vectors are observations from the original local datasets, and their number need not be small. Consequently, communicating support vectors may conflict with privacy or communication restrictions in distributed learning. Related approaches use coresets or other reduced subsets to decrease the amount of data needed for SVM training.

A second class of methods communicates model parameters and enforces agreement between local SVMs. For example, consensus-based distributed SVM can assign a local classifier $(\mathbf{w}_k, b_k)$ to each node and solve

$$\begin{aligned} \min_{\{\mathbf{w}_k, b_k\}_{k=1}^K} \quad & \sum_{k=1}^{K} \left[\frac{1}{2}\|\mathbf{w}_k\|_2^2 + C \sum_{i \in I_k} \max\left\{0, 1 - y_i(\mathbf{w}_k^\top \mathbf{x}_i + b_k)\right\} \right] \\ \text{subject to } & \mathbf{w}_1 = \cdots = \mathbf{w}_K, \\ & b_1 = \cdots = b_K. \end{aligned} \tag{4}$$

Such formulations can be solved by alternating local SVM optimization with consensus updates, for example through ADMM. Rather than exchanging local observations, communication is shifted to model parameters and consensus variables.

SVMs can also be incorporated into more general distributed or federated optimization frameworks. Local nodes may optimize the hinge loss using subgradient methods and periodically aggregate model parameters, while proximal penalties can be introduced to restrict local models from drifting too far from a shared model. These approaches avoid direct exchange of training observations, but their communication remains model-based. When local datasets are heterogeneous, multiple local optimization steps can move node-specific models in different directions before communication, creating the same local-drift issue that arises more generally in distributed optimization.

These approaches illustrate a broader design question in distributed SVMs: what information should be communicated so that decentralized optimization remains close to the centralized SVM objective? Support vector methods communicate selected observations, consensus and federated methods communicate model information, and coreset methods communicate reduced subsets of the data. Each choice determines which aspects of the local learning problem are preserved during communication. For SVMs, this question is particularly important because the hinge loss is nonsmooth and depends critically on the distribution of observations near the decision margin. A compact distributed representation should therefore capture not only the location of local observations but also the information relevant to their contribution to the SVM objective. This motivates the development of representative learning methods specifically adapted to distributed large-margin classification.

3 Representative Learning for Distributed SVM

We develop RepL methods for training SVMs from data distributed across heterogeneous nodes. Building on the Lifted ERM and clouded-representative framework introduced in Wang and Li (2026), we replace each point-valued covariate by a local probability distribution and evaluate the hinge loss through its expectation over that distribution. For Gaussian clouds, the resulting lifted hinge loss has a closed-form, differentiable expression that incorporates both the location and covariance of the cloud. This formulation enables gradient-based optimization and provides the foundation for constructing class-wise clouded representatives carrying local geometric and optimization information.

We first develop the method for linear SVMs. We derive the Gaussian lifted hinge loss, establish its principal analytical properties, and construct clouded MR and SMR. These representatives are communicated between nodes and incorporated into the local learning objectives to reduce the effects of data heterogeneity and local model drift without exchanging raw observations, gradients, or model parameters. We then extend the framework to kernel SVMs, where representatives lie in a reproducing kernel Hilbert space and require a common representation that can be communicated across nodes.

3.1 Linear SVM

3.1.1 Lifted SVM

To obtain a smooth objective while retaining information about local variability, we formulate the SVM through Lifted ERM, in which each point-valued covariate is replaced by a probability distribution. Specifically, we lift each observation $\mathbf{x}_i$ to a Gaussian cloud

$$\tilde{\mathbf{X}}_i \sim N(\mathbf{x}_i, \boldsymbol{\Sigma}), \tag{5}$$

where $\boldsymbol{\Sigma} = \operatorname{diag}\left(\tilde{\sigma}_1^2, \ldots, \tilde{\sigma}_p^2\right)$. A convenient default choice is the coordinate-wise rule-of-thumb bandwidth $\tilde{\sigma}_j = \widehat{\sigma}_j \left(\frac{3n}{4}\right)^{-1/5}$, $j = 1, \ldots, p$, with $\widehat{\sigma}_j$ the sample standard deviation of the jth covariate. This choice follows the classical rule-of-thumb bandwidth used in Gaussian density estimation. More generally, $\boldsymbol{\Sigma}$ controls the scale and geometry of the lifting and need not be restricted to a diagonal form.

Under the lifting in (5), the hinge-loss contribution of observation i becomes

$$\tilde{\ell}_i(\mathbf{w}, b) = \mathbb{E}_{\tilde{\mathbf{X}}_i}\left[\max\left\{0, 1 - y_i(\mathbf{w}^\top \tilde{\mathbf{X}}_i + b)\right\}\right]. \tag{6}$$

Consequently, the lifted SVM solves

$$\min_{\mathbf{w}, b} \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n} \tilde{\ell}_i(\mathbf{w}, b). \tag{7}$$

The expectation in (6) has an exact closed form. Define the random margin

$$M_i = y_i(\mathbf{w}^\top \tilde{\mathbf{X}}_i + b).$$

Since $\tilde{\mathbf{X}}_i$ is Gaussian,

$$M_i \sim N(\nu_i, s^2),$$

where

$$\nu_i = y_i(\mathbf{w}^\top \mathbf{x}_i + b), \quad s^2 = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}. \tag{8}$$

Let

$$a_i = \frac{1 - \nu_i}{s},$$

and denote the standard normal distribution and density functions by Φ and ϕ, respectively. For $s > 0$,

$$\tilde{\ell}_i(\mathbf{w}, b) = (1 - \nu_i)\Phi(a_i) + s\phi(a_i). \tag{9}$$

Thus, rather than replacing the hinge loss by an unrelated smooth surrogate, the lifted formulation evaluates the *exact expected hinge loss* under the Gaussian cloud.

Figure 1 illustrates the smoothing induced by Gaussian lifting around the nondifferentiable margin boundary.

For a single observation, write

$$\ell_\Sigma(\mathbf{w}, b) = (1 - \nu)\Phi(a) + s\phi(a). \tag{10}$$

Unless otherwise stated, assume $\boldsymbol{\Sigma} \succeq 0$. For results involving derivatives, we further restrict attention to $s \geq \epsilon > 0$.

Proposition 1 (Smoothness of the lifted hinge loss)**.** *Suppose that $s = \sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}} \geq \epsilon > 0$ on the parameter set under consideration and that $\mathbf{x}$ and $\boldsymbol{\Sigma}$ are bounded. Then $\ell_\Sigma(\mathbf{w}, b)$ has a Lipschitz-continuous gradient on this set. Consequently, ℓ_Σ is L-smooth for some finite $L > 0$.*

Proof. Differentiating (10) gives

$$\nabla_w \ell_\Sigma(\mathbf{w}, b) = -y\Phi(a)x + \phi(a)\frac{\boldsymbol{\Sigma}\mathbf{w}}{s}, \tag{11}$$

$$\frac{\partial \ell_\Sigma(\mathbf{w}, b)}{\partial b} = -y\Phi(a). \tag{12}$$

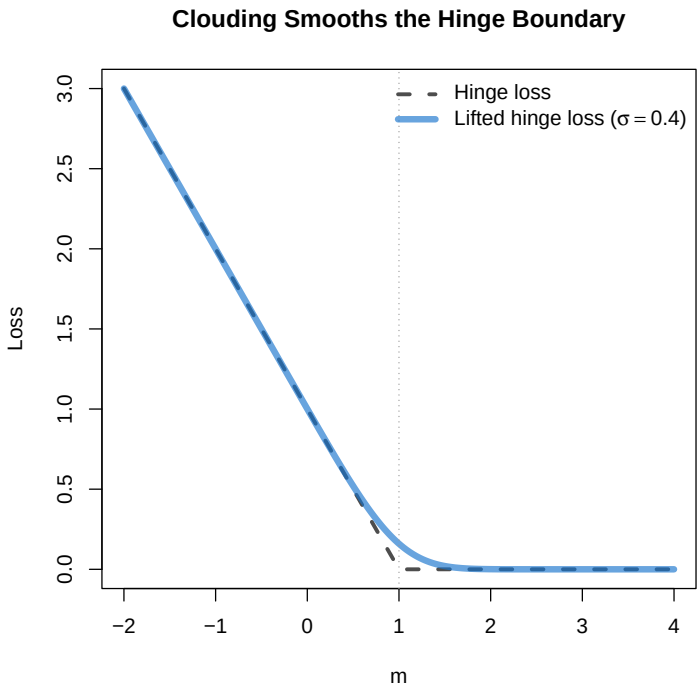


Figure 1: Hinge loss vs lifted hinge loss.

The corresponding second derivatives are

$$\nabla_w^2 \ell_\Sigma(\mathbf{w}, b) = \frac{\phi(a)}{s}\left(y\mathbf{x} + a\frac{\boldsymbol{\Sigma}\mathbf{w}}{s}\right)\left(y\mathbf{x} + a\frac{\boldsymbol{\Sigma}\mathbf{w}}{s}\right)^\top + \phi(a)\left\{\frac{\Sigma}{s} - \frac{(\boldsymbol{\Sigma}\mathbf{w})(\boldsymbol{\Sigma}\mathbf{w})^\top}{s^3}\right\}, \tag{13}$$

$$\nabla_w \frac{\partial \ell_\Sigma}{\partial b} = \frac{\phi(a)}{s}\left(y\mathbf{x} + a\frac{\boldsymbol{\Sigma}\mathbf{w}}{s}\right), \tag{14}$$

$$\frac{\partial^2 \ell_\Sigma}{\partial b^2} = \frac{\phi(a)}{s}. \tag{15}$$

Since $\phi(a) \le (2\pi)^{-1/2}$, $s \ge \epsilon > 0$, and the remaining quantities are bounded on the parameter set, the Hessian has bounded operator norm. Hence the gradient is Lipschitz continuous. □

Proposition 2 (Convexity and regularized strong convexity)**.** *The lifted hinge loss $\ell_\Sigma(\mathbf{w}, b)$ is convex in $(\mathbf{w}, b)$. Moreover, for $\lambda > 0$, adding $\lambda\|\mathbf{w}\|^2/2$ makes the resulting objective λ-strongly convex in $\mathbf{w}$ for each fixed b.*

Proof. Using the expectation representation,

$$\ell_\Sigma(\mathbf{w}, b) = \mathbb{E}_{\xi \sim N(0,\boldsymbol{\Sigma})}\left[\max\left\{0, 1 - y\big(\mathbf{w}^\top(\mathbf{x} + \xi) + b\big)\right\}\right].$$

For each fixed ξ, the expression inside the expectation is the maximum of two convex functions

of $(\mathbf{w}, b)$ and is therefore convex. Expectation preserves convexity, proving the first claim. Since $\lambda\|\mathbf{w}\|^2/2$ is λ-strongly convex in $\mathbf{w}$, adding it to the convex lifted loss establishes the second claim. □

Proposition 3 (Degeneration to the ordinary hinge loss)**.** *For fixed* $(\mathbf{w}, b, \mathbf{x})$*,*

$$\lim_{\mathbf{\Sigma}\to 0} \ell_\Sigma(\mathbf{w}, b) = \max\left\{0, 1 - y(\mathbf{w}^\top \mathbf{x} + b)\right\}. \tag{16}$$

More precisely,

$$\left|\ell_\Sigma(\mathbf{w}, b) - \max\left\{0, 1 - y(\mathbf{w}^\top \mathbf{x} + b)\right\}\right| \le \|\mathbf{w}\|\sqrt{\mathrm{tr}(\mathbf{\Sigma})}. \tag{17}$$

Proof. Let

$$h(z) = \max\{0, 1 - y(\mathbf{w}^\top \mathbf{z} + b)\}.$$

Since h is $\|\mathbf{w}\|$-Lipschitz,

$$|h(\mathbf{x} + \xi) - h(\mathbf{x})| \le \|\mathbf{w}\|\|\xi\|.$$

Therefore,

$$\begin{aligned} |\ell_\Sigma(\mathbf{w}, b) - h(\mathbf{x})| &\le \mathbb{E}|h(\mathbf{x} + \xi) - h(\mathbf{x})| \\ &\le \|\mathbf{w}\|\mathbb{E}\|\xi\| \\ &\le \|\mathbf{w}\|\sqrt{\mathbb{E}\|\xi\|^2} = \|\mathbf{w}\|\sqrt{\mathrm{tr}(\mathbf{\Sigma})}, \end{aligned}$$

where $\xi \sim N(0, \mathbf{\Sigma})$. The right-hand side converges to zero as $\mathbf{\Sigma} \to 0$, which proves both (17) and (16). □

The preceding result establishes convergence of the loss for fixed model parameters. The next result shows that, under standard identifiability and compactness conditions, the solutions of the lifted and ordinary SVM problems are also asymptotically equivalent.

Proposition 4 (Convergence of minimizers)**.** *Consider*

$$\mathcal{L}_\Sigma(\mathbf{w}, b) = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^n \ell_\Sigma^{(i)}(\mathbf{w}, b), \tag{18}$$

where

$$\ell_\Sigma^{(i)}(\mathbf{w}, b) = \mathbb{E}_{\xi_i \sim N(0,\mathbf{\Sigma})}\left[\max\left\{0, 1 - y_i\big(\mathbf{w}^\top(\mathbf{x}_i + \xi_i) + b\big)\right\}\right],$$

and let

$$\mathcal{L}_0(\mathbf{w}, b) = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^n \max\left\{0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)\right\}. \tag{19}$$

Assume that $\mathcal{L}_0$ *has a unique minimizer* $(\mathbf{w}_0, b_0)$ *and that the minimizers* $(\mathbf{w}_\Sigma, b_\Sigma)$ *of* $\mathcal{L}_\Sigma$ *remain in a common compact parameter set as* $\mathbf{\Sigma} \to 0$*. Then*

$$(\mathbf{w}_\Sigma, b_\Sigma) \longrightarrow (\mathbf{w}_0, b_0), \tag{20}$$

and

$$\mathcal{L}_\Sigma(\mathbf{w}_\Sigma, b_\Sigma) \to \mathcal{L}_0(\mathbf{w}_0, b_0). \tag{21}$$

Proof. By Proposition 3, for each observation,

$$\left|\ell_\Sigma^{(i)}(\mathbf{w}, b) - \max\{0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)\}\right| \le \|\mathbf{w}\| \sqrt{\operatorname{tr}(\mathbf{\Sigma})}.$$

Hence, on any compact parameter set,

$$|\mathcal{L}_\Sigma(\mathbf{w}, b) - \mathcal{L}_0(\mathbf{w}, b)| \le Cn\|\mathbf{w}\| \sqrt{\operatorname{tr}(\mathbf{\Sigma})} \longrightarrow 0 \tag{22}$$

uniformly as $\mathbf{\Sigma} \to 0$. Thus $\mathcal{L}_\Sigma \to \mathcal{L}_0$ uniformly on the common compact set containing the minimizers. Since $\mathcal{L}_0$ has a unique minimizer, standard argmin consistency yields $(\mathbf{w}_\Sigma, b_\Sigma) \to (\mathbf{w}_0, b_0)$. Uniform convergence then also implies convergence of the corresponding minimum objective values. □

Propositions 1-4 characterize the role of the Gaussian lifting from both optimization and approximation perspectives. For nondegenerate clouds, the lifted loss is smooth and convex, while quadratic regularization provides strong convexity in $\mathbf{w}$. This permits standard gradient-based optimization of the lifted objective. At the same time, the lifted loss and, under the stated conditions, its minimizer approach their ordinary SVM counterparts as $\mathbf{\Sigma} \to 0$. The differentiability of the nondegenerate lifted loss is especially useful for constructing representatives by matching local gradient information, which we develop next.

3.1.2 Clouded Representatives for SVM

Following the class structure of binary classification, representatives are constructed separately for each class on each node. Let $\mathcal{I}_{k,c}$ denote the indices of observations belonging to class $c \in \{-1, 1\}$ on node k, with $y_i = c$, $i \in \mathcal{I}_{k,c}$, and let $n_{k,c} = |\mathcal{I}_{k,c}|$. We represent this group by the clouded pseudo-observation

$$\tilde{\mathcal{D}}_{k,c} = \left(n_{k,c}, \tilde{y}_{k,c}, \tilde{\mathbf{x}}_{k,c}, \tilde{\mathbf{\Sigma}}_{k,c}\right), \tag{23}$$

where $\tilde{y}_{k,c} = c$. Thus, the representative preserves the class label and sample weight while replacing the original covariates by a representative location $\tilde{\mathbf{x}}_{k,c}$ and covariance $\tilde{\mathbf{\Sigma}}_{k,c}$.

For each observation $i \in \mathcal{I}_{k,c}$, define

$$\begin{aligned} \nu_i &= c(\mathbf{w}^\top \mathbf{x}_i + b), \\ s_{k,c} &= \sqrt{\mathbf{w}^\top \mathbf{\Sigma}_{k,c} w}, \\ a_i &= \frac{1 - \nu_i}{s_{k,c}}, \end{aligned}$$

where $\mathbf{\Sigma}_{k,c}$ is the covariance used to lift observations in class c on node k. For the clouded

representative, define correspondingly

$$\tilde{\nu}_{k,c} = c(\mathbf{w}^\top \tilde{\mathbf{x}}_{k,c} + b),$$
$$\tilde{s}_{k,c} = \sqrt{\mathbf{w}^\top \tilde{\mathbf{\Sigma}}_{k,c} w},$$
$$\tilde{a}_{k,c} = \frac{1 - \tilde{\nu}_{k,c}}{\tilde{s}_{k,c}}.$$

We consider two constructions that differ in the information they preserve: a model-free clouded MR and a model-dependent clouded SMR.

Clouded Mean Representative (MR). The simplest construction uses the class-wise covariate mean as the representative location,

$$\tilde{\mathbf{x}}_{k,c} = \frac{1}{n_{k,c}} \sum_{i \in \mathcal{I}_{k,c}} \mathbf{x}_i. \tag{24}$$

We retain the class-wise lifting covariance by setting

$$\tilde{\mathbf{\Sigma}}_{k,c} = \mathbf{\Sigma}_{k,c}. \tag{25}$$

Together with $\tilde{y}_{k,c} = c$, equations (24)-(25) define the clouded MR

$$\tilde{\mathcal{D}}^{\mathrm{MR}}_{k,c} = \left(n_{k,c}, c, \tilde{\mathbf{x}}_{k,c}, \mathbf{\Sigma}_{k,c}\right).$$

The representative location is therefore determined entirely by the local data and does not depend on the current SVM parameters. Clouded MR can consequently be interpreted as a distributional summary of the class-wise local data rather than a local approximation constructed for a particular model iterate.

Clouded Score-Matching Representative (SMR). Clouded MR summarizes the local data distribution but does not use the current learning objective. To incorporate optimization information, we construct a clouded Score-Matching Representative (SMR) by matching the gradient contribution of the observations in $\mathcal{I}_{k,c}$ with that of a single weighted clouded representative. In this context, "score" refers to the gradient of the lifted SVM objective.

From equations (11) and (12), matching the derivative with respect to b gives

$$\frac{1}{n_{k,c}} \sum_{i \in \mathcal{I}_{k,c}} \Phi(a_i) = \Phi(\tilde{a}_{k,c}). \tag{26}$$

Since Φ is strictly increasing,

$$\tilde{a}_{k,c} = \Phi^{-1}\left\{\frac{1}{n_{k,c}}\sum_{i\in\mathcal{I}_{k,c}}\Phi(a_i)\right\}. \tag{27}$$

Matching the gradient with respect to $\mathbf{w}$ gives

$$\frac{1}{n_{k,c}}\sum_{i\in\mathcal{I}_{k,c}}\left[-c\Phi(a_i)\mathbf{x}_i + \phi(a_i)\frac{\boldsymbol{\Sigma}_{k,c}w}{s_{k,c}}\right] = -c\Phi(\tilde{a}_{k,c})\tilde{\mathbf{x}}_{k,c} + \phi(\tilde{a}_{k,c})\frac{\tilde{\boldsymbol{\Sigma}}_{k,c}w}{\tilde{s}_{k,c}}. \tag{28}$$

Equations (26) and (28) require the clouded representative to reproduce both components of the class-wise average gradient at the parameter values $(\mathbf{w}, b)$ at which the representative is constructed.

To obtain a tractable covariance representation, we restrict the representative covariance to be proportional to the class-wise covariance,

$$\tilde{\boldsymbol{\Sigma}}_{k,c} = \delta_{\Sigma,k,c}\boldsymbol{\Sigma}_{k,c}, \tag{29}$$

where $\delta_{\Sigma,k,c} > 0$ is determined by the score-matching equations. This retains the covariance geometry of the local class while allowing its overall scale to adjust to match the lifted gradient.

Premultiplying (28) by $\mathbf{w}^\top$ gives

$$\mathcal{G}_{k,c} = -c\Phi(\tilde{a}_{k,c})\mathbf{w}^\top\tilde{\mathbf{x}}_{k,c} + \phi(\tilde{a}_{k,c})\tilde{s}_{k,c}, \tag{30}$$

where

$$\mathcal{G}_{k,c} = \frac{1}{n_{k,c}}\sum_{i\in\mathcal{I}_{k,c}}\left[-c\Phi(a_i)\mathbf{w}^\top\mathbf{x}_i + \phi(a_i)s_{k,c}\right]. \tag{31}$$

Using the definition of $\tilde{a}_{k,c}$,

$$\tilde{s}_{k,c} = \frac{1 - c(\mathbf{w}^\top\tilde{\mathbf{x}}_{k,c} + b)}{\tilde{a}_{k,c}}. \tag{32}$$

Substitution into (30) yields

$$\mathbf{w}^\top\tilde{\mathbf{x}}_{k,c} = \left[c\left\{\Phi(\tilde{a}_{k,c}) + \frac{\phi(\tilde{a}_{k,c})}{\tilde{a}_{k,c}}\right\}\right]^{-1} \times \left[(1-cb)\frac{\phi(\tilde{a}_{k,c})}{\tilde{a}_{k,c}} - \mathcal{G}_{k,c}\right]. \tag{33}$$

Once the projected location is obtained, $\tilde{s}_{k,c}$ follows from (32). From (29),

$$\delta_{\Sigma,k,c} = \frac{\tilde{s}_{k,c}^2}{\mathbf{w}^\top\boldsymbol{\Sigma}_{k,c}w}, \tag{34}$$

which determines $\tilde{\boldsymbol{\Sigma}}_{k,c}$.

Finally, rearranging the full gradient-matching equation (28) gives

$$\tilde{\mathbf{x}}_{k,c} = [c\Phi(\tilde{a}_{k,c})]^{-1} \left\{ \phi(\tilde{a}_{k,c}) \frac{\tilde{\mathbf{\Sigma}}_{k,c} w}{\tilde{s}_{k,c}} - \frac{1}{n_{k,c}} \sum_{i \in \mathcal{I}_{k,c}} \left[-c\Phi(a_i)\mathbf{x}_i + \phi(a_i) \frac{\mathbf{\Sigma}_{k,c} w}{s_{k,c}} \right] \right\}. \tag{35}$$

Unlike clouded MR, clouded SMR depends on the current parameters $(\mathbf{w}, b)$ and is therefore reconstructed as the model evolves. At its construction point, it preserves the class-wise gradient contribution of the local lifted SVM objective. Thus, MR provides a model-independent summary of local class location and spread, whereas SMR additionally encodes first-order information from the current lifted SVM objective.

3.1.3 Distributed Representative Learning

We now incorporate the clouded representatives into distributed SVM training. Let

$$\mathcal{R}_k = \{\tilde{\mathcal{D}}_{h,c} : h \neq k,\ c \in \{-1, 1\}\}$$

denote the representatives received by node k. Node k augments its local objective with the representatives received from the other nodes by minimizing the representative augmented lifted objective

$$\mathcal{L}_k(\mathbf{w}, b) = \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{i \in \mathcal{I}_k} \tilde{\ell}_i(\mathbf{w}, b) + C \sum_{h \neq k} \sum_{c \in \{-1,1\}} n_{h,c} \tilde{\ell}_{h,c}(\mathbf{w}, b), \tag{36}$$

where

$$\tilde{\ell}_{h,c}(\mathbf{w}, b) = (1 - \tilde{\nu}_{h,c})\Phi(\tilde{a}_{h,c}) + \tilde{s}_{h,c}\phi(\tilde{a}_{h,c}). \tag{37}$$

The weight $n_{h,c}$ accounts for the number of observations summarized by the corresponding representative. Thus, each remote class contributes to the local optimization through a single distribution-valued pseudo-observation rather than through its individual observations.

Representative exchange and local updates. Let $\theta_k^{(t)} = \left(\mathbf{w}_k^{(t)}, b_k^{(t)}\right)$ denote the parameter on node k at the beginning of communication round t. At each outer iteration, node k constructs one clouded representative for each locally observed class,

$$\tilde{\mathcal{D}}_{k,c}^{(t)} = \left(n_{k,c}, c, \tilde{\mathbf{x}}_{k,c}^{(t)}, \tilde{\mathbf{\Sigma}}_{k,c}^{(t)}\right), \quad c \in \{-1, 1\}. \tag{38}$$

For clouded MR, the representative is independent of the current model and therefore need not be reconstructed across communication rounds. For clouded SMR, the representative depends on the current parameter $\theta_k^{(t)}$ and is recomputed before each communication round.

The representatives are then communicated to the other participating nodes, either directly or

through a coordinating server. No raw observations, gradients, or model parameters are required to construct the representative-augmented objective on another node.

After receiving the representatives from the other nodes, node k forms the representative-augmented objective (36). Thus, the raw observations on node h are replaced on node k by their class-wise clouded representatives, with $n_{h,c}$ retaining the contribution of the number of observations summarized by each representative.

The representatives remain fixed during the local optimization phase. Starting from $\theta_{k,0}^{(t)} = \theta_k^{(t)}$, node k performs J gradient steps on (36),

$$\theta_{k,j+1}^{(t)} = \theta_{k,j}^{(t)} - \gamma_t \nabla \tilde{\mathcal{L}}_k^{(t)} \left(\theta_{k,j}^{(t)} \right), \tag{39}$$

for $j = 0, \ldots, J-1$, where $\theta_{k,j}^{(t)} = (\mathbf{w}_{k,j}^{(t)}, b_{k,j}^{(t)})$ and $\gamma_t > 0$ is the local step size. At the end of the local optimization phase,

$$\theta_k^{(t+1)} = \theta_{k,J}^{(t)}. \tag{40}$$

The procedure then repeats: model-dependent representatives are reconstructed at the updated parameters, communicated, and held fixed during the next sequence of local updates.

The number of local steps J controls the trade-off between local computation and communication frequency. When $J = 1$, representatives are refreshed after every gradient step. Larger J reduces the frequency of communication but allows the model to move farther from the parameter value at which a clouded SMR was constructed. This distinction is less consequential for clouded MR because its construction is independent of the current model.

Aggregation through representatives. Unlike parameter-aggregation methods, the proposed procedure does not average the local parameters $\theta_k^{(t)}$ after each communication round. Instead, information from the distributed data is aggregated through the representative-augmented objectives (36). Each node retains its own parameter iterate while optimizing against its local observations and the same collection of communicated clouded representatives. Consequently, agreement between local models, when it occurs, results from their optimization against shared representative information rather than from an explicit parameter-consensus step.

3.1.4 Theoretical Analysis

Clouded SMR matches the class-wise gradient of the lifted SVM objective at the parameter value used to construct the representative. Once constructed, however, the representative remains fixed during the subsequent local optimization steps. We therefore study how its location and covariance depend on within-group heterogeneity, how well the frozen representative approximates the full-data gradient away from its construction point, and how this approximation affects the optimization iterates.

Algorithm 1 Distributed Representative Learning for Linear SVM

Require: Local datasets $\{\mathcal{D}_k\}_{k=1}^K$, number of local steps J, step sizes $\{\gamma_t\}$, regularization parameter C.

Initialize $\theta_k^{(0)} = (\mathbf{w}_k^{(0)}, b_k^{(0)})$ for each node k.
for $t = 0, 1, \ldots, T-1$ **do**
  for all nodes k **do**
    for $c \in \{-1, 1\}$ **do**
      Construct $\tilde{\mathcal{D}}_{k,c}^{(t)}$ using clouded MR or clouded SMR.
    end for
    Communicate $\{\tilde{\mathcal{D}}_{k,c}^{(t)}\}_c$ to the participating nodes.
  end for
  for all nodes k **do**
    Receive $\{\tilde{\mathcal{D}}_{h,c}^{(t)} : h \neq k\}$.
    Set $\theta_{k,0}^{(t)} = \theta_k^{(t)}$.
    for $j = 0, \ldots, J-1$ **do**
      $\theta_{k,j+1}^{(t)} \leftarrow \theta_{k,j}^{(t)} - \gamma_t \nabla \tilde{\mathcal{L}}_k^{(t)}(\theta_{k,j}^{(t)})$.
    end for
    $\theta_k^{(t+1)} \leftarrow \theta_{k,J}^{(t)}$.
  end for
end for

Ensure: Final local models $\{\theta_k^{(T)}\}_{k=1}^K$.

For each node k and class $c \in \{-1, 1\}$, define

$$\mathbf{X}_{k,c} = \{\mathbf{x}_i : i \in \mathcal{I}_{k,c}\},$$
$$\Delta_{k,c} = \max_{i,j \in \mathcal{I}_{k,c}} \|\mathbf{x}_i - \mathbf{x}_j\|,$$
$$\Delta = \max_{k,c} \Delta_{k,c}.$$

Thus, $\Delta_{k,c}$ measures the within-group covariate diameter, while Δ is the largest diameter among the node-class groups.

We first examine a single group and suppress the indices (k, c). Let $\boldsymbol{\theta}_0 = (\mathbf{w}_0, b_0)$ denote the parameter at which the clouded SMR is constructed. Define the construction-point quantities

$$s_0 = \sqrt{\mathbf{w}_0^\top \boldsymbol{\Sigma} \mathbf{w}_0},$$
$$\nu_{i,0} = c(\mathbf{w}_0^\top \mathbf{x}_i + b_0),$$
$$a_{i,0} = \frac{1 - \nu_{i,0}}{s_0}.$$

The representative covariance is assumed to have the proportional form

$$\tilde{\boldsymbol{\Sigma}} = \delta_\Sigma \boldsymbol{\Sigma}, \tag{41}$$

where $\delta_\Sigma > 0$. Define

$$\begin{aligned}\tilde{s}_0 &= \sqrt{\mathbf{w}_0^\top \tilde{\mathbf{\Sigma}} \mathbf{w}_0},\\ \tilde{\nu}_0 &= c(\mathbf{w}_0^\top \tilde{\mathbf{x}} + b_0),\\ \tilde{a}_0 &= \frac{1-\tilde{\nu}_0}{\tilde{s}_0}.\end{aligned}$$

At $\boldsymbol{\theta}_0$, the score-matching conditions (26) and (28), with the group indices suppressed, become

$$\frac{1}{n}\sum_{i=1}^{n} \Phi(a_{i,0}) = \Phi(\tilde{a}_0), \tag{42}$$

$$\frac{1}{n}\sum_{i=1}^{n}\left[-c\Phi(a_{i,0})\mathbf{x}_i + \phi(a_{i,0})\frac{\mathbf{\Sigma}\mathbf{w}_0}{s_0}\right] = -c\Phi(\tilde{a}_0)\tilde{\mathbf{x}} + \phi(\tilde{a}_0)\frac{\tilde{\mathbf{\Sigma}}\mathbf{w}_0}{\tilde{s}_0}. \tag{43}$$

Lemma 1 (SMR covariance-scale stability)**.** *Let*

$$\mathcal{U}_r^{(0)} = \{(\mathbf{w}, b) : \|(\mathbf{w}, b) - (\mathbf{w}_0, b_0)\| \le r\}.$$

Suppose that $\mathbf{\Sigma}$ *is symmetric and positive definite and that there exist constants* $M_w, M_b, M_x, \epsilon_s > 0$ *such that, for all* $(\mathbf{w}, b) \in \mathcal{U}_r^{(0)}$*,*

$$\begin{aligned}\|\mathbf{w}\| &\le M_w,\\ |b| &\le M_b,\\ \max_i \|\mathbf{x}_i\| &\le M_x,\\ \sqrt{\mathbf{w}^\top \mathbf{\Sigma}\mathbf{w}} &\ge \epsilon_s.\end{aligned}$$

Define

$$\begin{aligned}A_r &= \frac{1 + M_w M_x + M_b}{\epsilon_s},\\ \mathcal{A}_r &= [-A_r, A_r],\\ m_{\phi,r} &= \inf_{a\in\mathcal{A}_r} \phi(a),\\ h_{\min,r} &= \inf_{a\in\mathcal{A}_r} h(a),\end{aligned}$$

where $h(a) = a\Phi(a) + \phi(a)$*. Then* $m_{\phi,r} > 0$*,* $h_{\min,r} > 0$*, and*

$$\sqrt{\delta_\Sigma} = \frac{n^{-1}\sum_{i=1}^{n} h(a_{i,0})}{h(\tilde{a}_0)}. \tag{44}$$

Moreover,

$$\left|\sqrt{\delta_\Sigma} - 1\right| \le \frac{M_w}{h_{\min,r}\epsilon_s}\left(1 + \frac{L_\Phi}{m_{\phi,r}}\right)\Delta, \tag{45}$$

where $L_\Phi = \sup_{a\in\mathbb{R}} \phi(a) = \frac{1}{\sqrt{2\pi}}$. *Consequently,* $\left|\sqrt{\delta_\Sigma} - 1\right| = \mathcal{O}(\Delta)$. *For* Δ *sufficiently small,* $|\delta_\Sigma - 1| = \mathcal{O}(\Delta)$.

Proof. From (41),

$$\tilde{s}_0 = \sqrt{\mathbf{w}_0^\top \delta_\Sigma \boldsymbol{\Sigma} \mathbf{w}_0} = \sqrt{\delta_\Sigma}\, s_0.$$

Premultiplying (43) by $\mathbf{w}_0^\top$ gives

$$-c\Phi(\tilde{a}_0)\mathbf{w}_0^\top \tilde{\mathbf{x}} + \phi(\tilde{a}_0)\tilde{s}_0 = \frac{1}{n}\sum_{i=1}^n \left[-c\Phi(a_{i,0})\mathbf{w}_0^\top \mathbf{x}_i + \phi(a_{i,0})s_0\right]. \tag{46}$$

From the definitions of the standardized margins,

$$c\mathbf{w}_0^\top \tilde{\mathbf{x}} = 1 - cb_0 - \tilde{a}_0\tilde{s}_0,$$
$$c\mathbf{w}_0^\top \mathbf{x}_i = 1 - cb_0 - a_{i,0}s_0.$$

Substituting these expressions into (46) and applying (42) cancels the terms containing $1 - cb_0$. We obtain

$$h(\tilde{a}_0)\tilde{s}_0 = s_0 \frac{1}{n}\sum_{i=1}^n h(a_{i,0}).$$

Since $\tilde{s}_0/s_0 = \sqrt{\delta_\Sigma}$, this proves (44).

The assumed parameter bounds imply

$$|a_{i,0}| = \frac{|1 - c(\mathbf{w}_0^\top \mathbf{x}_i + b_0)|}{s_0} \le A_r.$$

Therefore, $a_{i,0} \in \mathcal{A}_r$. By (42) and monotonicity of Φ, $\tilde{a}_0 \in \mathcal{A}_r$ as well. Hence $m_{\phi,r} > 0$. Also, $h'(a) = \Phi(a)$, and $h(a) > 0$ for every finite a, so continuity on the compact interval $\mathcal{A}_r$ gives $h_{\min,r} > 0$.

Let

$$\overline{a}_0 = \frac{1}{n}\sum_{i=1}^n a_{i,0},$$
$$\overline{h}_0 = \frac{1}{n}\sum_{i=1}^n h(a_{i,0}).$$

For any i, j,

$$|a_{i,0} - a_{j,0}| = \frac{|\mathbf{w}_0^\top (\mathbf{x}_i - \mathbf{x}_j)|}{s_0} \leq \frac{M_w}{\epsilon_s}\Delta.$$

It follows that

$$|a_{i,0} - \overline{a}_0| \leq \frac{M_w}{\epsilon_s}\Delta. \tag{47}$$

Because $h'(a) = \Phi(a)$ and $0 \leq \Phi(a) \leq 1$, h is 1-Lipschitz. Therefore,

$$\begin{aligned} |\overline{h}_0 - h(\overline{a}_0)| &\leq \frac{1}{n}\sum_{i=1}^{n} |a_{i,0} - \overline{a}_0| \\ &\leq \frac{M_w}{\epsilon_s}\Delta. \end{aligned} \tag{48}$$

Next, by (42),

$$\Phi(\tilde{a}_0) = \frac{1}{n}\sum_{i=1}^{n} \Phi(a_{i,0}).$$

Using the L_Φ-Lipschitz property of Φ and (47),

$$\begin{aligned} |\Phi(\tilde{a}_0) - \Phi(\overline{a}_0)| &\leq \frac{L_\Phi}{n}\sum_{i=1}^{n} |a_{i,0} - \overline{a}_0| \\ &\leq L_\Phi \frac{M_w}{\epsilon_s}\Delta. \end{aligned}$$

Since both arguments lie in $\mathcal{A}_r$, the mean value theorem and the definition of $m_{\phi,r}$ give

$$|\tilde{a}_0 - \overline{a}_0| \leq \frac{L_\Phi}{m_{\phi,r}} \frac{M_w}{\epsilon_s}\Delta. \tag{49}$$

Using again the 1-Lipschitz property of h,

$$|h(\tilde{a}_0) - h(\overline{a}_0)| \leq \frac{L_\Phi}{m_{\phi,r}} \frac{M_w}{\epsilon_s}\Delta. \tag{50}$$

By (44),

$$\left|\sqrt{\delta_\Sigma} - 1\right| = \frac{|\overline{h}_0 - h(\tilde{a}_0)|}{h(\tilde{a}_0)}.$$

Combining (48) and (50), and using $h(\tilde{a}_0) \geq h_{\min,r}$, proves (45).

Finally,

$$|\delta_\Sigma - 1| = \left|\sqrt{\delta_\Sigma} - 1\right| \left(\sqrt{\delta_\Sigma} + 1\right).$$

For sufficiently small Δ, $\sqrt{\delta_\Sigma}$ remains bounded, and hence $|\delta_\Sigma - 1| = \mathcal{O}(\Delta)$. □

Lemma 1 is a local stability result rather than a degenerating-covariance asymptotic result. For an empirical within-group covariance, $\|\widehat{\boldsymbol{\Sigma}}\|$ may itself decrease with the group diameter. In implementation, we stabilize the covariance through $\boldsymbol{\Sigma} = \widehat{\boldsymbol{\Sigma}} + \eta I$, where $\eta > 0$, so that the lifted standard deviation remains bounded away from zero.

We next compare the gradient evaluated on the full distributed data with the gradient evaluated on fixed clouded SMR representatives.

Theorem 1 (Gradient discrepancy for frozen clouded SMR)**.** *At a construction point $\boldsymbol{\theta}_{0,k} = (\mathbf{w}_{0,k}, b_{0,k})$, construct the clouded representative $\tilde{\mathbf{X}}_{k,c} = \left(\tilde{\mathbf{x}}_{k,c}, \tilde{\boldsymbol{\Sigma}}_{k,c}\right)$ satisfying the bias- and weight-matching equations, with $\tilde{\boldsymbol{\Sigma}}_{k,c} = \delta_{\Sigma,k,c}\boldsymbol{\Sigma}_{k,c}$.*

Suppose the assumptions of Lemma 1 hold uniformly over all groups. In particular, suppose

$$\left|\sqrt{\delta_{\Sigma,k,c}} - 1\right| \le C_\Sigma \Delta_{k,c} \tag{51}$$

for a constant C_Σ independent of (k, c).

For $r > 0$, define

$$\mathcal{U}_{k,r} = \{\boldsymbol{\theta} : \|\boldsymbol{\theta} - \boldsymbol{\theta}_{0,k}\| \le r\},$$

and suppose $\mathcal{U}_r = \bigcap_k \mathcal{U}_{k,r}$ is nonempty. Assume uniformly over all groups and all $\boldsymbol{\theta} = (\mathbf{w}, b) \in \mathcal{U}_{k,r}$ that

$$\begin{aligned}
\|\mathbf{w}\| &\le M_w, \\
|b| &\le M_b, \\
\|\mathbf{x}_i\| &\le M_x, \\
\|\boldsymbol{\Sigma}_{k,c}\| &\le M_\Sigma, \\
s_{k,c}(\mathbf{w}) = \sqrt{\mathbf{w}^\top \boldsymbol{\Sigma}_{k,c}\mathbf{w}} &\ge \epsilon_s, \\
\tilde{s}_{k,c}(\mathbf{w}) = \sqrt{\mathbf{w}^\top \tilde{\boldsymbol{\Sigma}}_{k,c}\mathbf{w}} &\ge \epsilon_s.
\end{aligned}$$

Assume also that the original and representative standardized margins remain in a common compact interval.

Define the full-data and frozen-representative objectives by

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{k,c}\alpha_{k,c}\ell_c(\mathbf{X}_{k,c};\boldsymbol{\theta}), \tag{52}$$

$$\tilde{\mathcal{L}}(\boldsymbol{\theta}) = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{k,c}\alpha_{k,c}\ell_c(\tilde{\mathbf{X}}_{k,c};\boldsymbol{\theta}), \tag{53}$$

where $\alpha_{k,c} \geq 0$, $\sum_{k,c}\alpha_{k,c} = 1$. *Then there exists a constant* $\kappa > 0$, *independent of* Δ, *such that*

$$\sup_{\boldsymbol{\theta}\in\mathcal{U}_r}\left\|\nabla\mathcal{L}(\boldsymbol{\theta}) - \nabla\tilde{\mathcal{L}}(\boldsymbol{\theta})\right\| \leq \kappa\Delta. \tag{54}$$

Proof. Fix a group (k, c) and suppress its indices. Let

$$\overline{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i.$$

We first obtain structural bounds on the representative at the construction point. Combining the weight-matching and bias-matching equations gives

$$\begin{aligned}\Phi(\tilde{a}_0)\,(\tilde{\mathbf{x}} - \overline{\mathbf{x}}) = \frac{1}{n}\sum_{i=1}^{n}\Phi(a_{i,0})\,(\mathbf{x}_i - \overline{\mathbf{x}}) \\ + c\left[\phi(\tilde{a}_0)\frac{\tilde{\boldsymbol{\Sigma}}\mathbf{w}_0}{\tilde{s}_0} - \left\{\frac{1}{n}\sum_{i=1}^{n}\phi(a_{i,0})\right\}\frac{\boldsymbol{\Sigma}\mathbf{w}_0}{s_0}\right].\end{aligned} \tag{55}$$

Because $\overline{\mathbf{x}}$ belongs to the convex hull of the group, $\|\mathbf{x}_i - \overline{\mathbf{x}}\| \leq \Delta$. Therefore,

$$\left\|\frac{1}{n}\sum_{i=1}^{n}\Phi(a_{i,0})(\mathbf{x}_i - \overline{\mathbf{x}})\right\| \leq \Delta. \tag{56}$$

Since $\tilde{\boldsymbol{\Sigma}} = \delta_\Sigma\boldsymbol{\Sigma}$,

$$\frac{\tilde{\boldsymbol{\Sigma}}\mathbf{w}_0}{\tilde{s}_0} = \sqrt{\delta_\Sigma}\frac{\boldsymbol{\Sigma}\mathbf{w}_0}{s_0}.$$

Lemma 1, together with the Lipschitz continuity of ϕ on the compact standardized-margin interval, implies

$$\left|\phi(\tilde{a}_0)\sqrt{\delta_\Sigma} - \frac{1}{n}\sum_{i=1}^{n}\phi(a_{i,0})\right| \leq C_1\Delta$$

for a constant $C_1 > 0$. Moreover,

$$\left\|\frac{\boldsymbol{\Sigma}\mathbf{w}_0}{s_0}\right\| \leq \frac{M_\Sigma M_w}{\epsilon_s}.$$

Since $\tilde{a}_0$ remains in a compact interval, $\Phi(\tilde{a}_0)$ is bounded below by a positive constant. Applying these bounds to (55) yields

$$\|\tilde{\mathbf{x}} - \overline{\mathbf{x}}\| \leq C_x \Delta \tag{57}$$

for some $C_x > 0$. Consequently,

$$\|\mathbf{x}_i - \tilde{\mathbf{x}}\| \leq (1 + C_x)\Delta. \tag{58}$$

Now fix an arbitrary $\boldsymbol{\theta} = (\mathbf{w}, b) \in \mathcal{U}_{k,r}$. The representative remains fixed, so the matching equations are not assumed to hold at this new parameter.

By (51),

$$|s - \tilde{s}| = \left|1 - \sqrt{\delta_\Sigma}\right| s \leq C_s \Delta \tag{59}$$

for some $C_s > 0$. Similarly, (58) gives

$$|\nu_i - \tilde{\nu}| \leq \|\mathbf{w}\| \|\mathbf{x}_i - \tilde{\mathbf{x}}\| \leq C_\nu \Delta. \tag{60}$$

Because the model parameters and representative location are bounded, $|1 - \tilde{\nu}|$ is uniformly bounded. Since $s, \tilde{s} \geq \epsilon_s$,

$$\begin{aligned} |a_i - \tilde{a}| &\leq \frac{|\nu_i - \tilde{\nu}|}{s} + |1 - \tilde{\nu}| \frac{|s - \tilde{s}|}{s\tilde{s}} \\ &\leq C_a \Delta \end{aligned}$$

for some $C_a > 0$.

The bias derivatives satisfy

$$\begin{aligned} \frac{\partial}{\partial b} \ell_c(\mathbf{X}; \boldsymbol{\theta}) &= -\frac{c}{n} \sum_{i=1}^{n} \Phi(a_i), \\ \frac{\partial}{\partial b} \ell_c(\tilde{\mathbf{X}}; \boldsymbol{\theta}) &= -c\Phi(\tilde{a}). \end{aligned}$$

Since Φ is globally Lipschitz,

$$\left| \frac{\partial}{\partial b} \ell_c(\mathbf{X}; \boldsymbol{\theta}) - \frac{\partial}{\partial b} \ell_c(\tilde{\mathbf{X}}; \boldsymbol{\theta}) \right| \leq C_b \Delta. \tag{61}$$

For the weight-gradient location component,

$$\begin{aligned}
&\|\Phi(a_i)\mathbf{x}_i - \Phi(\tilde{a})\tilde{\mathbf{x}}\| \\
\leq &|\Phi(a_i) - \Phi(\tilde{a})|\|\mathbf{x}_i\| + \Phi(\tilde{a})\|\mathbf{x}_i - \tilde{\mathbf{x}}\| \\
\leq &C_{\text{loc}}\Delta.
\end{aligned}$$

Averaging over i gives the same order.

For the covariance component, add and subtract $\phi(\tilde{a})\boldsymbol{\Sigma}\mathbf{w}/s$. The Lipschitz continuity of ϕ on the compact standardized-margin interval gives

$$\left|\frac{1}{n}\sum_{i=1}^{n}\phi(a_i) - \phi(\tilde{a})\right| \leq C_\phi\Delta.$$

Furthermore,

$$\frac{\tilde{\boldsymbol{\Sigma}}\mathbf{w}}{\tilde{s}} = \sqrt{\delta_\Sigma}\frac{\boldsymbol{\Sigma}\mathbf{w}}{s},$$

and hence

$$\left\|\frac{\tilde{\boldsymbol{\Sigma}}\mathbf{w}}{\tilde{s}} - \frac{\boldsymbol{\Sigma}\mathbf{w}}{s}\right\| \leq C_{\text{cov}}\Delta.$$

Combining the location and covariance terms yields

$$\left\|\nabla_{\mathbf{w}}\ell_c(\mathbf{X};\boldsymbol{\theta}) - \nabla_{\mathbf{w}}\ell_c(\tilde{\mathbf{X}};\boldsymbol{\theta})\right\| \leq C_w\Delta. \tag{62}$$

Together with (61), this implies

$$\sup_{\boldsymbol{\theta}\in\mathcal{U}_{k,r}}\left\|\nabla\ell_c(\mathbf{X}_{k,c};\boldsymbol{\theta}) - \nabla\ell_c(\tilde{\mathbf{X}}_{k,c};\boldsymbol{\theta})\right\| \leq C_g\Delta_{k,c}. \tag{63}$$

Finally, the regularization gradients in (52) and (53) are identical and cancel. Therefore,

$$\begin{aligned}
&\left\|\nabla\mathcal{L}(\boldsymbol{\theta}) - \nabla\tilde{\mathcal{L}}(\boldsymbol{\theta})\right\| \\
\leq &C\sum_{k,c}\alpha_{k,c}\left\|\nabla\ell_c(\mathbf{X}_{k,c};\boldsymbol{\theta}) - \nabla\ell_c(\tilde{\mathbf{X}}_{k,c};\boldsymbol{\theta})\right\| \\
\leq &CC_g\sum_{k,c}\alpha_{k,c}\Delta_{k,c} \\
\leq &CC_g\Delta.
\end{aligned}$$

Taking the supremum over $\mathcal{U}_r$ and setting $\kappa = CC_g$ proves (54).

□

Theorem 1 describes the approximation quality of a frozen clouded SMR. Exact gradient matching holds at the construction point, while the discrepancy over a nearby optimization region is controlled to first order by the largest within-group diameter. Thus, class-wise subgroups with compact covariate geometry produce more faithful gradient representations.

We conclude by translating the gradient-discrepancy bound into an optimization guarantee for the iterative representative-learning procedure.

Corollary 1 (Inexact gradient descent with iterative clouded SMR)**.** *Let $\mathcal{L} : \mathbb{R}^{p+1} \to \mathbb{R}$ denote the full-data objective, and let $\boldsymbol{\theta}^*$ be its minimizer. Suppose that $\mathcal{L}$ is μ-strongly convex and L-smooth on a convex neighborhood $\mathcal{U}$.*

At outer iteration t, construct clouded SMR representatives at $\boldsymbol{\theta}^{(t)}$, producing the frozen representative objective $\tilde{\mathcal{L}}^{(t)}$. Set $\boldsymbol{\theta}_0^{(t)} = \boldsymbol{\theta}^{(t)}$ and perform J inner updates

$$\boldsymbol{\theta}_{j+1}^{(t)} = \boldsymbol{\theta}_j^{(t)} - \gamma \nabla \tilde{\mathcal{L}}^{(t)} \left(\boldsymbol{\theta}_j^{(t)} \right), \tag{64}$$

where $j = 0, \ldots, J-1$. Define $\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}_J^{(t)}$. Assume that all outer and inner iterates remain in $\mathcal{U}$, and that

$$\sup_{\boldsymbol{\theta} \in \mathcal{U}} \left\| \nabla \tilde{\mathcal{L}}^{(t)}(\boldsymbol{\theta}) - \nabla \mathcal{L}(\boldsymbol{\theta}) \right\| \leq \kappa \Delta \tag{65}$$

uniformly over t. If $0 < \gamma \leq \frac{1}{L}$, and $\rho = 1 - \gamma\mu$, then after T outer iterations,

$$\|\boldsymbol{\theta}^{(T)} - \boldsymbol{\theta}^*\| \leq \rho^{JT} \|\boldsymbol{\theta}^{(0)} - \boldsymbol{\theta}^*\| + \frac{\kappa \Delta}{\mu} \left(1 - \rho^{JT}\right). \tag{66}$$

Consequently,

$$\limsup_{T \to \infty} \|\boldsymbol{\theta}^{(T)} - \boldsymbol{\theta}^*\| \leq \frac{\kappa \Delta}{\mu}. \tag{67}$$

Proof. For each outer iteration t and inner iteration j, define

$$\mathbf{e}_{t,j} = \nabla \tilde{\mathcal{L}}^{(t)} \left(\boldsymbol{\theta}_j^{(t)} \right) - \nabla \mathcal{L} \left(\boldsymbol{\theta}_j^{(t)} \right).$$

The update in (64) can be written as

$$\boldsymbol{\theta}_{j+1}^{(t)} = \boldsymbol{\theta}_j^{(t)} - \gamma \nabla \mathcal{L} \left(\boldsymbol{\theta}_j^{(t)} \right) - \gamma \mathbf{e}_{t,j}.$$

Since $\mathcal{L}$ is μ-strongly convex and L-smooth, and $0 < \gamma \leq 1/L$, the exact gradient map is contractive:

$$\begin{aligned} &\left\| \boldsymbol{\theta}_j^{(t)} - \gamma \nabla \mathcal{L} \left(\boldsymbol{\theta}_j^{(t)} \right) - \boldsymbol{\theta}^* \right\| \\ \leq &\rho \left\| \boldsymbol{\theta}_j^{(t)} - \boldsymbol{\theta}^* \right\|. \end{aligned}$$

Therefore,

$$\begin{aligned}\left\|\boldsymbol{\theta}_{j+1}^{(t)}-\boldsymbol{\theta}^{*}\right\| &\leq \rho\left\|\boldsymbol{\theta}_{j}^{(t)}-\boldsymbol{\theta}^{*}\right\|+\gamma\|\mathbf{e}_{t,j}\| \\ &\leq \rho\left\|\boldsymbol{\theta}_{j}^{(t)}-\boldsymbol{\theta}^{*}\right\|+\gamma\kappa\Delta.\end{aligned}$$

Iterating over the J inner updates gives

$$\begin{aligned}\left\|\boldsymbol{\theta}^{(t+1)}-\boldsymbol{\theta}^{*}\right\| &\leq \rho^{J}\left\|\boldsymbol{\theta}^{(t)}-\boldsymbol{\theta}^{*}\right\|+\gamma\kappa\Delta\sum_{j=0}^{J-1}\rho^{j} \\ &= \rho^{J}\left\|\boldsymbol{\theta}^{(t)}-\boldsymbol{\theta}^{*}\right\|+\frac{\kappa\Delta}{\mu}\left(1-\rho^{J}\right).\end{aligned}$$

Applying this recursion over T outer iterations proves (66). Taking the limit superior gives (67). □

The strong-convexity assumption in Corollary 1 applies to the full parameter $\boldsymbol{\theta}=(\mathbf{w},b)$. The standard SVM regularizer guarantees strong convexity in $\mathbf{w}$ but does not regularize the intercept b. The corollary therefore applies directly when the intercept is fixed, eliminated, or regularized. Without such a condition, the corresponding contraction guarantee should be interpreted with respect to the regularized weight parameter.

Remark 1 (Exact first step after reconstruction). Suppose that all clouded SMR representatives are constructed at the same parameter $\boldsymbol{\theta}^{(t)}$. By score matching,

$$\nabla\widetilde{\mathcal{L}}^{(t)}\left(\boldsymbol{\theta}^{(t)}\right)=\nabla\mathcal{L}\left(\boldsymbol{\theta}^{(t)}\right).$$

Hence the first inner step after each representative reconstruction has zero gradient-approximation error. Repeating the proof of Corollary 1 over the remaining $J-1$ potentially inexact steps gives

$$\left\|\boldsymbol{\theta}^{(t+1)}-\boldsymbol{\theta}^{*}\right\| \leq \rho^{J}\left\|\boldsymbol{\theta}^{(t)}-\boldsymbol{\theta}^{*}\right\|+\frac{\kappa\Delta}{\mu}\left(1-\rho^{J-1}\right).$$

In particular, when $J=1$, the representative objective produces the same gradient update as the full-data objective at every outer iteration.

3.2 Kernel SVM

The distributed problem becomes more involved for nonlinear SVMs. Given a feature map $\psi:\mathbb{R}^{p}\to\mathcal{H}$ into a reproducing kernel Hilbert space (RKHS) $\mathcal{H}$, a kernel $K(\mathbf{x},\mathbf{x}')$ satisfies

$$K(\mathbf{x},\mathbf{x}')=\left\langle\psi(\mathbf{x}),\psi(\mathbf{x}')\right\rangle_{\mathcal{H}}.$$

The representer theorem yields a finite expansion of the learned function in terms of kernel evaluations at observed data points. Consequently, nonlinear SVM training introduces additional

storage and computational costs associated with the kernel representation. Low-dimensional or low-rank approximations, including random Fourier features and Nyström approximations, provide one route for reducing these costs and can facilitate distributed implementations.

For the lifted formulation, an additional issue arises from the variance term. In the linear setting, Gaussian lifting leads to $s^2 = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$, which is the variance of the linear score induced by the cloud distribution. To obtain the corresponding nonlinear formulation, let

$$f(\mathbf{x}) = \langle w, \psi(\mathbf{x}) \rangle_{\mathcal{H}} + b,$$

and let Γ denote the covariance operator associated with $\psi(X)$,

$$\Gamma = \mathbb{E}\left[(\psi(X) - \mu_\psi) \otimes (\psi(X) - \mu_\psi) \right], \tag{68}$$

where $\mu_\psi = \mathbb{E}[\psi(X)]$. Then

$$\mathrm{Var}\{f(X)\} = \langle w, \Gamma w \rangle_{\mathcal{H}}. \tag{69}$$

Thus, $\langle w, \Gamma w \rangle_{\mathcal{H}}$ plays the role of $\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$ in the linear lifted SVM.

Directly communicating an empirical covariance operator is generally unsuitable for distributed representative learning because the local RKHS representations may be expressed in terms of different local observations. We therefore introduce a shared dictionary $\mathcal{B} = \{\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_m\}$, available to all participating nodes. The dictionary serves both as a finite-dimensional model space and as a common coordinate system in which clouded representatives can be communicated.

Define the dictionary Gram matrix $\mathbf{K} \in \mathbb{R}^{m \times m}$ and the dictionary kernel vector $\mathbf{k}(\mathbf{x}) \in \mathbb{R}^m$ by

$$[\mathbf{K}]_{jl} = K(\boldsymbol{\xi}_j, \boldsymbol{\xi}_l), \tag{70}$$

$$\mathbf{k}(\mathbf{x}) = \left(K(\boldsymbol{\xi}_1, \mathbf{x}), \ldots, K(\boldsymbol{\xi}_m, \mathbf{x}) \right)^\top. \tag{71}$$

We restrict the classifier to the dictionary span,

$$w = \sum_{j=1}^{m} \beta_j \psi(\boldsymbol{\xi}_j), \tag{72}$$

so that

$$f(\mathbf{x}) = \boldsymbol{\beta}^\top \mathbf{k}(\mathbf{x}) + b, \quad \|w\|_{\mathcal{H}}^2 = \boldsymbol{\beta}^\top \mathbf{K} \boldsymbol{\beta}.$$

For a class-wise group $\mathcal{D}_{k,c} = \{\mathbf{x}_i : i \in \mathcal{I}_{k,c}\}$, let

$$\mathbf{K}_{k,c} = [\mathbf{k}(\mathbf{x}_i)]_{i \in \mathcal{I}_{k,c}} \in \mathbb{R}^{m \times n_{k,c}},$$

and let

$$\mathbf{H}_{k,c} = \mathbf{I}_{n_{k,c}} - \frac{1}{n_{k,c}} \mathbf{1}\mathbf{1}^\top.$$

We define the dictionary-based covariance matrix

$$\mathbf{C}_{k,c} = \frac{1}{n_{k,c}} \mathbf{K}_{k,c} \mathbf{H}_{k,c} \mathbf{K}_{k,c}^\top. \tag{73}$$

For a parameter $\boldsymbol{\beta}$, the corresponding lifted scale is

$$s_{k,c} = \sqrt{\boldsymbol{\beta}^\top \mathbf{C}_{k,c} \boldsymbol{\beta}}. \tag{74}$$

Equivalently,

$$s_{k,c}^2 = \frac{1}{n_{k,c}} \left(\mathbf{K}_{k,c}^\top \boldsymbol{\beta} \right)^\top \mathbf{H}_{k,c} \left(\mathbf{K}_{k,c}^\top \boldsymbol{\beta} \right),$$

so $s_{k,c}^2$ is the empirical variance of the kernel scores within the node-class group. This provides a finite-dimensional analogue of (69) that is computable using kernel evaluations.

For $i \in \mathcal{I}_{k,c}$, define

$$\nu_i = c \left(\boldsymbol{\beta}^\top \mathbf{k}(\mathbf{x}_i) + b \right),$$
$$a_i = \frac{1 - \nu_i}{s_{k,c}}.$$

The kernel lifted hinge loss retains the same form as in the linear case,

$$\ell_i(\boldsymbol{\beta}, b) = (1 - \nu_i)\Phi(a_i) + s_{k,c}\phi(a_i), \tag{75}$$

with regularization

$$\frac{1}{2}\|w\|_{\mathcal{H}}^2 = \frac{1}{2}\boldsymbol{\beta}^\top \mathbf{K} \boldsymbol{\beta}. \tag{76}$$

Hence the lifted SVM framework extends to the nonlinear setting while remaining finite-dimensional with respect to the shared dictionary.

3.2.1 Clouded Representatives for Kernel SVM

We next construct class-wise clouded representatives in the dictionary coordinate system. For a group (k, c), let a representative be summarized by

$$\tilde{\mathcal{D}}_{k,c} = \left(n_{k,c}, c, \boldsymbol{\gamma}_{k,c}, \tilde{\mathbf{C}}_{k,c} \right), \tag{77}$$

where $\boldsymbol{\gamma}_{k,c} \in \mathbb{R}^m$ determines a representative in the dictionary span,

$$\tilde{z}_{k,c} = \sum_{j=1}^{m} \gamma_{k,c,j} \psi(\boldsymbol{\xi}_j).$$

Its dictionary interaction vector is

$$\tilde{\mathbf{r}}_{k,c} = \mathbf{K}\boldsymbol{\gamma}_{k,c}, \tag{78}$$

so that

$$\langle w, \tilde{z}_{k,c} \rangle_{\mathcal{H}} = \boldsymbol{\beta}^\top \tilde{\mathbf{r}}_{k,c}.$$

The representative margin and lifted scale are therefore

$$\begin{aligned}
\tilde{\nu}_{k,c} &= c\left(\boldsymbol{\beta}^\top \tilde{\mathbf{r}}_{k,c} + b\right), \\
\tilde{s}_{k,c} &= \sqrt{\boldsymbol{\beta}^\top \tilde{\mathbf{C}}_{k,c} \boldsymbol{\beta}}, \\
\tilde{a}_{k,c} &= \frac{1 - \tilde{\nu}_{k,c}}{\tilde{s}_{k,c}}.
\end{aligned}$$

Kernel Mean Representative. The analogue of the class-wise mean representative is the empirical kernel mean. Its dictionary interaction vector is

$$\mathbf{r}_{k,c}^{\mathrm{MR}} = \frac{1}{n_{k,c}} \sum_{i \in \mathcal{I}_{k,c}} \mathbf{k}(\mathbf{x}_i). \tag{79}$$

A dictionary-span realization is obtained by choosing $\boldsymbol{\gamma}_{k,c}^{\mathrm{MR}}$ to satisfy

$$\mathbf{K}\boldsymbol{\gamma}_{k,c}^{\mathrm{MR}} = \mathbf{r}_{k,c}^{\mathrm{MR}}. \tag{80}$$

When numerical regularization is required, the coefficients may instead be obtained from

$$(\mathbf{K} + \lambda_K \mathbf{I})\, \boldsymbol{\gamma}_{k,c}^{\mathrm{MR}} = \mathbf{r}_{k,c}^{\mathrm{MR}}, \quad \lambda_K > 0. \tag{81}$$

Kernel Score-Matching Representative. The kernel SMR is constructed by matching the class-wise lifted gradient. From (75), the derivatives of a single lifted observation are

$$\frac{\partial \ell_i}{\partial b} = -c\Phi(a_i), \tag{82}$$

$$\nabla_{\boldsymbol{\beta}} \ell_i = -c\Phi(a_i)\mathbf{k}(\mathbf{x}_i) + \phi(a_i)\frac{\mathbf{C}_{k,c}\boldsymbol{\beta}}{s_{k,c}}. \tag{83}$$

Matching the intercept derivative gives

$$\Phi(\tilde{a}_{k,c}) = \frac{1}{n_{k,c}} \sum_{i \in \mathcal{I}_{k,c}} \Phi(a_i), \tag{84}$$

and hence

$$\tilde{a}_{k,c} = \Phi^{-1} \left\{ \frac{1}{n_{k,c}} \sum_{i \in \mathcal{I}_{k,c}} \Phi(a_i) \right\}. \tag{85}$$

As in the linear construction, restrict the representative covariance to be proportional to the group covariance,

$$\tilde{\mathbf{C}}_{k,c} = \delta_{C,k,c} \mathbf{C}_{k,c}, \quad \delta_{C,k,c} > 0. \tag{86}$$

Then

$$\tilde{s}_{k,c} = \sqrt{\delta_{C,k,c}}\, s_{k,c}.$$

Define

$$\mathbf{p}_{k,c} = \frac{1}{n_{k,c}} \sum_{i \in \mathcal{I}_{k,c}} \Phi(a_i) \mathbf{k}(\mathbf{x}_i),$$
$$\bar{\phi}_{k,c} = \frac{1}{n_{k,c}} \sum_{i \in \mathcal{I}_{k,c}} \phi(a_i).$$

Matching the $\boldsymbol{\beta}$-gradient gives

$$-c\Phi(\tilde{a}_{k,c}) \tilde{\mathbf{r}}_{k,c} + \phi(\tilde{a}_{k,c}) \sqrt{\delta_{C,k,c}} \frac{\mathbf{C}_{k,c}\boldsymbol{\beta}}{s_{k,c}} = -c\mathbf{p}_{k,c} + \bar{\phi}_{k,c} \frac{\mathbf{C}_{k,c}\boldsymbol{\beta}}{s_{k,c}}. \tag{87}$$

Therefore,

$$\tilde{\mathbf{r}}_{k,c} = \frac{\mathbf{p}_{k,c}}{\Phi(\tilde{a}_{k,c})} + \frac{c}{\Phi(\tilde{a}_{k,c})} \left[\phi(\tilde{a}_{k,c}) \sqrt{\delta_{C,k,c}} - \bar{\phi}_{k,c} \right] \frac{\mathbf{C}_{k,c}\boldsymbol{\beta}}{s_{k,c}}. \tag{88}$$

The covariance scale follows from the same projected score-matching identity as in the linear case. Define $h(a) = a\Phi(a) + \phi(a)$. Then

$$\sqrt{\delta_{C,k,c}} = \frac{n_{k,c}^{-1} \sum_{i \in \mathcal{I}_{k,c}} h(a_i)}{h(\tilde{a}_{k,c})}, \tag{89}$$

or equivalently

$$\delta_{C,k,c} = \left[\frac{n_{k,c}^{-1} \sum_{i \in \mathcal{I}_{k,c}} h(a_i)}{h(\tilde{a}_{k,c})} \right]^2. \tag{90}$$

Finally, the representative coefficients are obtained from

$$(\mathbf{K} + \lambda_K \mathbf{I})\, \boldsymbol{\gamma}_{k,c}^{\mathrm{SMR}} = \tilde{\mathbf{r}}_{k,c}. \tag{91}$$

Thus, rather than communicating an element of the RKHS directly, node k communicates the finite-dimensional tuple

$$\left(n_{k,c}, c, \boldsymbol{\gamma}_{k,c}, \tilde{\mathbf{C}}_{k,c} \right).$$

Because all nodes use the same dictionary, the received representative can be evaluated under a recipient's current parameter $\boldsymbol{\beta}$ through $\boldsymbol{\beta}^\top \mathbf{K} \boldsymbol{\gamma}_{k,c}$ and $\boldsymbol{\beta}^\top \tilde{\mathbf{C}}_{k,c} \boldsymbol{\beta}$.

3.2.2 Distributed Representative Learning

The distributed kernel procedure follows the same representative-exchange structure as the linear case. At communication round t, node k constructs class-wise kernel representatives

$$\tilde{\mathcal{D}}_{k,c}^{(t)} = \left(n_{k,c}, c, \boldsymbol{\gamma}_{k,c}^{(t)}, \tilde{\mathbf{C}}_{k,c}^{(t)} \right)$$

and communicates them to the other participating nodes. Kernel MR representatives are model-independent, whereas kernel SMR representatives depend on the current coefficient vector and are reconstructed as the model evolves.

After receiving the representatives, node k minimizes the representative-augmented kernel objective

$$\tilde{\mathcal{L}}_k^{(t)}(\boldsymbol{\beta}, b) = \frac{1}{2} \boldsymbol{\beta}^\top \mathbf{K} \boldsymbol{\beta} + \frac{C}{n} \Bigg[\sum_{c \in \{-1,1\}} \sum_{i \in \mathcal{I}_{k,c}} \ell_{i,k}(\boldsymbol{\beta}, b) + \sum_{h \neq k} \sum_{c \in \{-1,1\}} n_{h,c} \tilde{\ell} \left(\tilde{\mathcal{D}}_{h,c}^{(t)}; \boldsymbol{\beta}, b \right) \Bigg], \tag{92}$$

where the lifted loss of a received representative is

$$\tilde{\ell} \left(\tilde{\mathcal{D}}_{h,c}^{(t)}; \boldsymbol{\beta}, b \right) = \left(1 - \tilde{\nu}_{h,c}^{(t)} \right) \Phi \left(\tilde{a}_{h,c}^{(t)} \right) + \tilde{s}_{h,c}^{(t)} \phi \left(\tilde{a}_{h,c}^{(t)} \right), \tag{93}$$

with

$$\tilde{\nu}_{h,c}^{(t)} = c\left(\boldsymbol{\beta}^\top \mathbf{K}\boldsymbol{\gamma}_{h,c}^{(t)} + b\right),$$
$$\tilde{s}_{h,c}^{(t)} = \sqrt{\boldsymbol{\beta}^\top \tilde{\mathbf{C}}_{h,c}^{(t)}\boldsymbol{\beta}},$$
$$\tilde{a}_{h,c}^{(t)} = \frac{1-\tilde{\nu}_{h,c}^{(t)}}{\tilde{s}_{h,c}^{(t)}}.$$

The representatives remain fixed during the local optimization phase, and node k performs updates in $(\boldsymbol{\beta}, b)$ analogously to the linear procedure. Consequently, the extension from the linear to the kernel setting preserves the same communication structure: nodes exchange weighted class-wise clouded representatives rather than raw observations or model parameters. The shared dictionary provides the additional common coordinate system required to express and evaluate these representatives across nodes.

4 Numerical Experiments

4.1 Linear SVM Experiments

4.1.1 Effect of Lifting

We first isolate the effect of Gaussian lifting from the effect of representative sharing. Specifically, we compare optimization of the ordinary hinge-loss objective with optimization of the lifted hinge-loss objective on the same single-node datasets. This comparison addresses two questions: whether lifting materially changes the resulting classification performance and whether the smooth lifted objective provides more favorable behavior under comparable first-order optimization methods.

For the ordinary hinge-loss objective, we use subgradient descent with the decreasing step-size schedule

$$\eta_t = \frac{\eta_0}{1+t}. \tag{94}$$

The initial step size is selected from $\eta_0 \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3\}$ separately for each simulation setting. The lifted objective is optimized using gradient descent with a fixed step size. These choices are intended to compare the two objectives using broadly comparable first-order procedures rather than specialized SVM solvers.

For each method, optimization performance is measured through the normalized best-so-far objective gap

$$G_t = \frac{\min\limits_{0\le s\le t} F(\boldsymbol{\theta}^{(s)}) - F^*}{F(\boldsymbol{\theta}^{(0)}) - F^*}, \tag{95}$$

where F denotes the objective being optimized and F^* is its minimum value. Because the ordinary

Table 1: Optimization and classification performance of ordinary and lifted hinge-loss objectives. Iterations report the number of first-order updates required to reach a normalized objective gap below 10^{-3}.

Regime	Objective	Iterations to 10^{-3}	Final accuracy
Standard	Lifted hinge	22.5	0.9353
	Ordinary hinge	81.4	0.9356
Few balanced	Lifted hinge	24.9	0.9648
	Ordinary hinge	130.1	0.9648
Heavy imbalance	Lifted hinge	46.2	0.9765
	Ordinary hinge	81.4	0.9771

and lifted losses define different objectives, the normalized gap is computed relative to the optimum of the corresponding objective. We record the number of iterations required to reach $G_t \leq 10^{-3}$, together with the final classification accuracy.

The experiments consider three broad simulation regimes: standard balanced samples, balanced samples with fewer observations, and samples with substantial class imbalance. Within these regimes, the data configurations vary in class separation and covariate dispersion. Table 1 summarizes the resulting optimization and classification performance.

The ordinary and lifted hinge-loss objectives yield nearly identical final classification accuracies in all three reported regimes. Thus, within these simulations, replacing the pointwise hinge loss by its Gaussian-lifted counterpart does not materially alter predictive performance. This observation is consistent with the limiting result in Proposition 3, although the simulations use nondegenerate lifting covariances.

The principal difference appears in optimization. Under the first-order methods considered here, gradient descent on the smooth lifted objective reaches the specified objective-gap tolerance in fewer iterations than subgradient descent on the ordinary hinge-loss objective. The difference is largest in the few-observation balanced regime, where the lifted objective requires 24.9 iterations on average, compared with 130.1 for the ordinary hinge objective.

This comparison should not be interpreted as establishing that gradient descent on the lifted objective is uniformly faster than specialized SVM optimization methods. Rather, it illustrates the computational effect of replacing a nonsmooth objective by a differentiable expected loss when comparable first-order procedures are used. The smooth gradient structure is particularly relevant to the clouded SMR construction, which uses local gradient information to construct and update representatives.

4.1.2 Distributed Learning under Heterogeneity

For the distributed experiments that follow, we use 100 independent replications for each configuration. Each dataset contains 600 observations in dimensions $d \in \{2, 3, 5, 7\}$, with equal global proportions of the two classes before distributed partitioning. The class-conditional distributions

are three-component Gaussian mixtures. For each class $c \in \{-1, 1\}$, the mixture weights are sampled according to $(\pi_{c,1}, \pi_{c,2}, \pi_{c,3}) \sim \text{Dirichlet}(1, 1, 1)$, and the component means are generated as $\boldsymbol{\mu}_{c,j} = \boldsymbol{\mu}_c + r_{c,j}\mathbf{z}_{c,j}$, $r_{c,j} \sim \text{Unif}(0, 0.75)$, where $\mathbf{z}_{c,j}$ is a randomly generated unit vector. Let m denote one-half of the distance between the two base class centers. The high-overlap configurations use $m \sim \text{Unif}(0.25, 1.00)$, whereas the low-overlap configurations use $m \sim \text{Unif}(1.00, 2.50)$.

The Gaussian-component covariance matrices are selected independently for the two classes from a covariance bank containing dispersion levels 0.25, 0.50, 1.00, 1.50, 2.00. At each dispersion level, the bank contains isotropic, mildly anisotropic, and strongly anisotropic covariance structures. We write the generating covariance as

$$\boldsymbol{\Omega}_c = s_c^2 \mathbf{A}_c, \quad \frac{\text{tr}(\mathbf{A}_c)}{d} = 1,$$

so that

$$\sqrt{\frac{\text{tr}(\boldsymbol{\Omega}_c)}{d}} = s_c.$$

For the isotropic structure, $\mathbf{A}_c = \mathbf{I}_d$. Before trace normalization, the mildly and strongly anisotropic structures assign, respectively, two and twenty-five times as much variance to the first coordinate. The generating covariances for the two classes are selected independently. Independent test observations are generated from the same class-conditional Gaussian-mixture distributions as the corresponding training observations.

The observations are distributed across three nodes using independent class-wise Dirichlet-multinomial allocations with concentration parameter $\alpha = 0.1$. The allocation is redrawn until every node contains at least two observations from each class. This procedure preserves global class balance while producing strongly heterogeneous local class proportions.

For each node-class group (k, c), the lifting covariance is estimated from the corresponding local observations. Let $\widehat{\sigma}_{k,c,j}$ denote the empirical standard deviation of the jth covariate among the $n_{k,c}$ observations in that group. We use the bandwidth factor

$$h_{k,c} = \left(\frac{3n_{k,c}}{4}\right)^{-1/5}$$

and define

$$\boldsymbol{\Sigma}_{k,c} = \text{diag}\left[\left(h_{k,c}\widehat{\sigma}_{k,c,1}\right)^2, \ldots, \left(h_{k,c}\widehat{\sigma}_{k,c,d}\right)^2\right]. \tag{96}$$

When this covariance cannot be estimated reliably, it is replaced by a small positive-definite diagonal matrix to maintain numerical stability.

All distributed methods use the same initialization, $\mathbf{w}^{(0)} = \mathbf{0}$, $b^{(0)} = 0$, followed by 20 local warm-start iterations before the first communication round. Unless otherwise stated, the distributed methods use learning rate 0.005, regularization parameter $C = 75$ with the loss normalized by the

Table 2: Test accuracy over 100 simulations for high-overlap classes under the default communication setting. Values are mean ± standard deviation.

Method	$d = 2$	$d = 3$	$d = 5$	$d = 7$
MR (Hinge)	0.7599 ± 0.0982	0.7811 ± 0.0920	0.7866 ± 0.0968	0.7834 ± 0.0913
SMR (Hinge)	0.6653 ± 0.1474	0.6838 ± 0.1497	0.6872 ± 0.1544	0.6746 ± 0.1391
SMR (Lifted)	0.7545 ± 0.1049	0.7860 ± 0.0910	$\mathbf{0.7995 \pm 0.1052}$	0.7879 ± 0.0965
MR (Lifted)	$\mathbf{0.7601 \pm 0.1003}$	$\mathbf{0.7873 \pm 0.0913}$	0.7990 ± 0.1039	$\mathbf{0.7938 \pm 0.0942}$
FedAvg (Lifted)	0.7345 ± 0.1258	0.7478 ± 0.1221	0.7650 ± 0.1241	0.7417 ± 0.1289
FedProx (Lifted)	0.7356 ± 0.1191	0.7469 ± 0.1251	0.7653 ± 0.1252	0.7427 ± 0.1266
ADMM (Hinge)	0.7437 ± 0.1223	0.7675 ± 0.1124	0.7833 ± 0.1308	0.7819 ± 0.1017
Classical SVM (Hinge)	0.7482 ± 0.1126	0.7861 ± 0.0933	0.7981 ± 0.1044	0.7911 ± 0.0952
Global Pooled (Lifted)	0.7539 ± 0.1044	0.7868 ± 0.0909	$\mathbf{0.8005 \pm 0.1057}$	0.7895 ± 0.0950

total sample size, 40 local optimization steps per communication round, and 25 communication rounds.

We consider both low- and high-overlap regimes. Since the high-overlap setting provides the more difficult classification problem and the stronger test of optimization under heterogeneous local class proportions, we report its predictive-accuracy results below. Both regimes are retained in the subsequent comparison of recovery of the centralized solution.

We compare the proposed clouded representatives with parameter-based and centralized alternatives. MR (Lifted) and SMR (Lifted) use the lifted hinge loss developed in Section 3.1.2. Their nonsmooth counterparts, MR (Hinge) and SMR (Hinge), use the ordinary hinge objective. We additionally include FedAvg and FedProx using the lifted loss, together with a distributed ADMM solution for the ordinary hinge objective. Two centralized benchmarks are included: Classical Linear SVM, which fits the ordinary hinge-loss objective to the pooled data, and Global Pooled (Lifted), which minimizes the lifted objective using all observations.

Under the high-overlap setting, the effect of heterogeneous local class proportions is visible in the parameter-aggregation methods. With 40 local updates between communications, FedAvg and FedProx have lower test accuracy than the centralized lifted solution across all four dimensions. In contrast, the two lifted representative methods remain close to the centralized benchmarks. In particular, MR (Lifted) and SMR (Lifted) produce test accuracies comparable to Global Pooled (Lifted) throughout the reported dimensions.

The comparison between SMR (Hinge) and SMR (Lifted) is also notable. Score matching based directly on the nonsmooth hinge objective performs substantially worse in these experiments, whereas clouded SMR constructed from the smooth lifted objective remains competitive with the centralized solutions. The corresponding difference between MR (Hinge) and MR (Lifted) is considerably smaller. This distinction is consistent with the role of lifting in the two constructions: MR is primarily a model-independent summary of class-wise location, while SMR directly uses objective-gradient information when constructing its representatives.

Table 3: Percentage of high-overlap simulations in which the local lifted-SMR models agree under different communication regimes.

Communication regime	$d = 2$	$d = 3$	$d = 5$	$d = 7$
Moderate	99%	100%	100%	95%
Single communication	0%	0%	0%	0%
Frequent	100%	100%	100%	96%

4.1.3 Communication Frequency

The preceding experiment uses 40 local gradient steps between representative exchanges. We next vary the communication frequency to examine how long communicated information remains useful as the local models move away from the parameter values at which it was constructed. We focus on the high-overlap setting, where local class imbalance and overlap provide the stronger test of distributed optimization.

We consider three communication regimes. The *moderate* regime uses 25 communication rounds with 40 local gradient steps per round. The *single-communication* regime performs only one representative exchange before extensive subsequent local optimization. Finally, the *frequent-communication* regime takes one local gradient step between successive communication rounds. These settings distinguish methods that depend strongly on the current model state from summaries that remain fixed throughout optimization.

In addition to predictive accuracy, we examine whether the local models agree after distributed training. For a given simulation, model agreement is declared when the maximum Euclidean parameter distance between any two nodes is less than 10^{-6}. Table 3 reports the percentage of simulations satisfying this criterion for lifted SMR.

The communication regimes reveal a clear distinction between MR and SMR. MR is determined by the local data and is independent of the current model parameter. Its representative therefore remains unchanged as local optimization proceeds. SMR instead matches first-order information at the parameter value where the representative is constructed. When many local updates occur without reconstruction, the local parameters can move away from this construction point and the frozen representative becomes a less accurate description of the current local gradient.

This behavior is most apparent in the single-communication regime. Under moderate communication, lifted SMR achieves model agreement in 95%-100% of the simulations across the four dimensions, and under frequent communication the corresponding values are 96%-100%. When the representatives are communicated only once, however, agreement falls to zero in all four dimensions. Thus, the SMR should be interpreted as a locally accurate, periodically refreshed representation of the optimization objective rather than as a static summary of the local data.

When communication is frequent and only one local update is performed between representative reconstructions, the distinction between distributed methods is correspondingly reduced. This behavior is also consistent with the exact score-matching property discussed in Remark 1: immediately after an SMR reconstruction, the first representative-gradient update agrees with the corresponding

Table 4: Euclidean distance $\|\mathbf{w} - \mathbf{w}_{\text{global}}\|_2$ from the Global Pooled (Lifted) solution under the default communication setting.

Setting	Method	$d = 2$	$d = 3$	$d = 5$	$d = 7$
Low overlap	SMR (Lifted)	**0.0444**	**0.0457**	**0.0618**	**0.0651**
	SMR (Hinge)	0.4497	0.5059	0.5156	0.5498
	MR (Lifted)	0.6032	0.5946	0.5951	0.5983
	MR (Hinge)	0.7943	0.7335	0.7337	0.7578
High overlap	SMR (Lifted)	**0.0475**	**0.0485**	**0.0699**	**0.0714**
	SMR (Hinge)	0.5200	0.5906	0.5872	0.6229
	MR (Lifted)	0.6860	0.6658	0.6641	0.6632
	MR (Hinge)	0.8847	0.8094	0.8059	0.8312

full-data gradient update at the construction parameter. The empirical results therefore illustrate the communication-local-optimization trade-off predicted by the frozen-representative analysis.

4.1.4 Recovery of the Centralized Solution

Predictive accuracy and agreement among nodes do not, by themselves, establish that a distributed method has recovered the centralized optimizer. Different linear classifiers can produce similar predictions while having substantially different parameter values. We therefore compare the representative-based solutions directly with the Global Pooled (Lifted) solution.

This comparison uses the default moderate-communication setting. For each method, we measure the Euclidean discrepancy between the learned weight vector and the centralized lifted-SVM weight vector. Table 4 reports the resulting distances for both the low- and high-overlap regimes. The corresponding comparisons of intercept discrepancy and cosine similarity lead to the same qualitative conclusion and are therefore omitted for brevity.

The parameter-distance comparison separates the roles of MR and SMR more clearly than predictive accuracy alone. Across every reported dimension and in both overlap regimes, lifted SMR produces the smallest discrepancy from the centralized lifted-SVM weight vector. Its distances range from 0.0444 to 0.0651 in the low-overlap setting and from 0.0475 to 0.0714 in the high-overlap setting. By comparison, the lifted-MR distances are approximately 0.59-0.60 under low overlap and 0.66-0.69 under high overlap.

This distinction helps explain why MR and SMR can have similar classification accuracy while behaving differently as optimization procedures. MR preserves class-wise data location and can therefore provide a strong predictive summary in these approximately linear settings, but it is not constructed to reproduce the local objective gradient. SMR is explicitly constructed to match that first-order information. The substantially smaller parameter discrepancies for lifted SMR therefore provide direct empirical evidence that score matching more faithfully recovers the optimizer of the centralized lifted objective.

Taken together, the three distributed experiments distinguish predictive performance, communication sensitivity, and optimization fidelity. Clouded MR provides a stable model-independent

summary that can remain useful under infrequent communication, while clouded SMR is model-dependent and benefits from periodic reconstruction. When refreshed during optimization, lifted SMR not only maintains competitive predictive accuracy under heterogeneous local class distributions but also most closely reproduces the centralized lifted-SVM solution.

4.2 Kernel SVM Experiments

We next evaluate the kernel extension on two binary-classification datasets using the Gaussian RBF kernel. In contrast to the synthetic experiments used for the linear SVM, these experiments use the Ionosphere and Australian Credit Approval datasets and evaluate the representative methods under heterogeneous class allocation across nodes.

4.2.1 Experimental Setup

We evaluate the kernel extension on the Ionosphere and Australian Credit Approval binary-classification datasets using the Gaussian RBF kernel. All reported results are averaged over 100 independent repetitions.

For each repetition, the observations are randomly divided into 85% training data and 15% test data. Covariate standardization is fitted using only the training data, and the resulting transformation is then applied to the private training observations, the shared dictionary, and the held-out test observations. Thus, the test data are not used to estimate the preprocessing transformation.

The kernel representative construction requires a common dictionary through which all nodes express their models and representatives. After standardization, a stratified random sample containing 10% of the training observations is selected as the shared dictionary. These observations are removed from the private training pool before the remaining observations are distributed across the nodes.

The Ionosphere dataset contains 351 observations, yielding 298 training observations and 53 test observations. Its shared dictionary contains 30 observations. The Australian Credit Approval dataset contains 690 observations, yielding 586 training observations and 104 test observations. Its shared dictionary contains 59 observations.

After removing the dictionary observations, the remaining private training data are distributed across three nodes using independent class-wise Dirichlet allocations with concentration parameter $\alpha = 0.1$. The allocation is redrawn until each node contains at least four observations in total and at least two observations from each class.

All kernel representative methods use the Gaussian RBF kernel

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2\right), \quad \gamma = \frac{1}{d}, \tag{97}$$

where d is the covariate dimension of the corresponding dataset. The distributed lifted methods use regularization parameter $C = 100$, with the empirical-loss term normalized by the total sample size.

The centralized classical kernel-SVM baseline uses $C = 1$.

Before the first representative exchange, each node performs 50 local warm-start optimization steps with learning rate 0.001. For the hinge-loss first-order methods, we use initial step size 0.1 together with an inverse-square-root step-size schedule.

As in the linear experiments, we compare three communication regimes. We denote by J/T a regime with J local optimization steps per communication round and T communication rounds, and consider 1/1000, 40/25, 1000/1. These correspond, respectively, to frequent communication, moderate communication, and a single communication round followed by extensive local optimization.

We compare lifted MR and SMR with their hinge-loss counterparts, FedAvg and FedProx using the lifted objective, and distributed ADMM using the ordinary hinge loss. We additionally report three centralized reference solutions. Kernel SVM (Hinge) fits the ordinary RBF-kernel SVM. Central Full (Lifted) minimizes the lifted kernel objective using the full training-set dictionary, whereas Central Shared (Lifted) minimizes the lifted objective in the smaller dictionary span available to the distributed methods. Central Shared (Lifted) therefore provides the most direct centralized target for evaluating the distributed representative methods.

4.2.2 Distributed Learning and Communication Frequency

Table 5 reports test accuracy under the three communication regimes. Because the distributed kernel methods are restricted to the common dictionary span, both centralized full-dictionary and shared-dictionary results are included. Their difference separates the approximation introduced by restricting the kernel model to the shared dictionary from the additional approximation introduced by distributed learning.

The centralized comparisons show that the effect of restricting the model to the shared dictionary depends on the dataset. On Ionosphere, the Central Shared Dictionary solution has test accuracy 0.9309, compared with 0.9383 for the full-dictionary lifted solution. On the Australian dataset, the corresponding accuracies are 0.8671 and 0.8668, respectively. Thus, the shared-dictionary restriction produces a visible accuracy difference for Ionosphere but essentially no difference for Australian in these experiments.

Under frequent communication, the primary distributed lifted methods are close to the Central Shared Dictionary solution on both datasets. As the number of local updates between communications increases, however, the parameter-aggregation methods become substantially more sensitive to local drift. For example, on Ionosphere, FedAvg decreases from 0.9275 under the 1/1000 regime to 0.8855 under 40/25 and 0.8408 under 1000/1. The corresponding Australian accuracies are 0.8658, 0.8158, and 0.7917.

Lifted MR and lifted SMR are considerably less sensitive to the moderate-communication regime. On Ionosphere, their test accuracies under 40/25 are 0.9233 and 0.9277, respectively, compared with 0.9309 for the centralized shared-dictionary objective. On Australian, both representative methods obtain 0.8657, compared with 0.8671 for the centralized shared-dictionary solution.

The single-communication regime also illustrates the different roles of MR and SMR. MR is

Table 5: Test accuracy over 100 repetitions for the kernel SVM experiments. Columns denote local steps per communication round / number of communication rounds. Values are mean ± standard deviation.

Dataset	Method	1/1000	40/25	1000/1
Ionosphere	MR (Hinge)	0.8550 ± 0.0541	0.8550 ± 0.0541	0.8550 ± 0.0541
	SMR (Hinge)	0.9189 ± 0.0398	0.9032 ± 0.0499	0.7473 ± 0.1640
	SMR (Lifted)	$\mathbf{0.9276 \pm 0.0373}$	$\mathbf{0.9277 \pm 0.0376}$	0.9101 ± 0.0486
	MR (Lifted)	0.9233 ± 0.0391	0.9233 ± 0.0391	$\mathbf{0.9233 \pm 0.0391}$
	FedAvg (Lifted)	0.9275 ± 0.0372	0.8855 ± 0.0810	0.8408 ± 0.1180
	FedProx (Lifted)	0.9275 ± 0.0372	0.8847 ± 0.0819	0.8387 ± 0.1168
	ADMM (Hinge)	0.9200 ± 0.0392	0.9204 ± 0.0401	0.7843 ± 0.1454
	Kernel SVM (Hinge)		0.9402 ± 0.0325	
	Central Full (Lifted)		0.9383 ± 0.0324	
	Central Shared (Lifted)		0.9309 ± 0.0342	
Australian	MR (Hinge)	0.8547 ± 0.0300	0.8547 ± 0.0300	0.8547 ± 0.0300
	SMR (Hinge)	0.8572 ± 0.0287	0.8424 ± 0.0442	0.7178 ± 0.1686
	SMR (Lifted)	$\mathbf{0.8658 \pm 0.0313}$	$\mathbf{0.8657 \pm 0.0312}$	0.8605 ± 0.0319
	MR (Lifted)	0.8657 ± 0.0322	$\mathbf{0.8657 \pm 0.0322}$	$\mathbf{0.8657 \pm 0.0322}$
	FedAvg (Lifted)	$\mathbf{0.8658 \pm 0.0314}$	0.8158 ± 0.0876	0.7917 ± 0.1137
	FedProx (Lifted)	$\mathbf{0.8658 \pm 0.0314}$	0.8151 ± 0.0893	0.7964 ± 0.1066
	ADMM (Hinge)	0.8618 ± 0.0330	0.8617 ± 0.0323	0.7264 ± 0.1532
	Kernel SVM (Hinge)		0.8586 ± 0.0310	
	Central Full (Lifted)		0.8668 ± 0.0308	
	Central Shared (Lifted)		0.8671 ± 0.0308	

Table 6: Percentage of repetitions in which the local lifted-SMR kernel models agree within RKHS-distance tolerance 10^{-2}.

Dataset	1/1000	40/25	1000/1
Ionosphere	90%	84%	0%
Australian	93%	93%	0%

independent of the current model and therefore remains unchanged as local optimization proceeds. SMR instead matches the objective gradient at its construction point and becomes less representative as the model moves away from that point without reconstruction. Accordingly, lifted MR is unchanged across the three communication regimes, while lifted SMR exhibits some degradation under 1000/1. Nevertheless, lifted SMR remains substantially more stable in this regime than both FedAvg and the hinge-based SMR construction.

To further examine the effect of representative refresh frequency, we measure agreement among the local lifted-SMR models. Agreement is defined using an RKHS-distance tolerance of 10^{-2}.

Lifted SMR achieves agreement in 84%–93% of repetitions when representatives are refreshed during training, but no repetitions satisfy the agreement criterion when only one representative exchange is performed. This behavior parallels the linear experiments and reinforces the interpretation of SMR as a local, model-dependent approximation that benefits from periodic reconstruction.

Table 7: RKHS distance from the centralized shared-dictionary lifted solution. Values are mean ± standard deviation.

Dataset	Method	1/1000	40/25	1000/1
Ionosphere	MR (Hinge)	2.0588 ± 0.2334	2.0588 ± 0.2334	2.0588 ± 0.2334
	MR (Lifted)	0.7246 ± 0.1988	0.7246 ± 0.1988	0.7246 ± 0.1988
	SMR (Hinge)	0.7684 ± 0.2062	0.9086 ± 0.2593	2.0234 ± 0.7213
	SMR (Lifted)	$\mathbf{0.5961 \pm 0.2039}$	$\mathbf{0.5960 \pm 0.2038}$	$\mathbf{0.6972 \pm 0.2386}$
Australian	MR (Hinge)	1.9901 ± 0.1043	1.9901 ± 0.1043	1.9901 ± 0.1043
	MR (Lifted)	0.5755 ± 0.0422	0.5755 ± 0.0422	0.5755 ± 0.0422
	SMR (Hinge)	0.6300 ± 0.1177	0.8669 ± 0.1938	1.8893 ± 0.6440
	SMR (Lifted)	$\mathbf{0.2967 \pm 0.0667}$	$\mathbf{0.2970 \pm 0.0666}$	$\mathbf{0.4642 \pm 0.1331}$

4.2.3 Recovery of the Centralized Solution

Classification accuracy does not determine whether a distributed classifier has recovered the centralized optimization solution. We therefore compare each representative-based model directly with the Central Shared Dictionary (Lifted) solution. Because all models are expressed in the same dictionary span, the discrepancy can be measured using the RKHS norm.

Let $w = \Psi_{\mathcal{B}}\boldsymbol{\beta}$, $w_g = \Psi_{\mathcal{B}}\boldsymbol{\beta}_g$ denote a distributed model and the corresponding centralized shared-dictionary model. Their RKHS distance is

$$\|w - w_g\|_{\mathcal{H}} = \sqrt{(\boldsymbol{\beta} - \boldsymbol{\beta}_g)^\top \mathbf{K}(\boldsymbol{\beta} - \boldsymbol{\beta}_g)}. \tag{98}$$

For each repetition, the distance is first averaged across the three local node models and then averaged across repetitions.

The RKHS-distance results provide a sharper distinction between MR and SMR than predictive accuracy alone. Lifted MR and lifted SMR have very similar test accuracy in several settings, but lifted SMR is consistently closer to the centralized shared-dictionary solution in RKHS distance. For example, under the moderate 40/25 regime, the mean RKHS distance on Ionosphere is 0.5960 for lifted SMR and 0.7246 for lifted MR. On Australian, the corresponding distances are 0.2970 and 0.5755.

This distinction is consistent with the representative constructions. MR summarizes class-wise location without depending on the current model, whereas SMR is explicitly constructed to reproduce first-order information from the lifted objective. Strong predictive performance from MR therefore does not necessarily imply recovery of the centralized optimizer. The RKHS-distance results show that lifted SMR more faithfully reproduces the centralized shared-dictionary kernel solution across all three communication regimes considered here.

4.2.4 Random Fourier Feature Approximation

As an alternative to the shared-dictionary kernel construction, we also consider a Random Fourier Feature (RFF) approximation to the RBF kernel. Whereas the shared-dictionary approach restricts

the classifier to a finite-dimensional subspace generated by selected dictionary observations, RFF provides all nodes with the same finite-dimensional Euclidean feature map. The linear lifted-SVM representative constructions can then be applied directly in this common feature space.

For the RBF kernel

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2\right),$$

we sample

$$\boldsymbol{\omega}_1, \ldots, \boldsymbol{\omega}_D \sim \mathcal{N}(\mathbf{0}, 2\gamma \mathbf{I})$$

and use the feature map

$$\mathbf{z}(\mathbf{x}) = \frac{1}{\sqrt{D}} \begin{pmatrix} \cos\left(\boldsymbol{\omega}_1^\top \mathbf{x}\right) \\ \sin\left(\boldsymbol{\omega}_1^\top \mathbf{x}\right) \\ \vdots \\ \cos\left(\boldsymbol{\omega}_D^\top \mathbf{x}\right) \\ \sin\left(\boldsymbol{\omega}_D^\top \mathbf{x}\right) \end{pmatrix}.$$

Then

$$\mathbf{z}(\mathbf{x})^\top \mathbf{z}(\mathbf{x}') \approx K(\mathbf{x}, \mathbf{x}'),$$

and the resulting classifier has the linear form

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{z}(\mathbf{x}) + b.$$

We use $D = 50$ sampled frequencies, producing a 100-dimensional feature representation, and otherwise follow the same three-node heterogeneous partitioning and communication regimes as in the shared-dictionary kernel experiments.

5 Conclusion and Future Work

5.1 Conclusion

We developed a Representative Learning framework for distributed support vector machines based on Lifted Empirical Risk Minimization. Rather than optimizing the nonsmooth hinge loss directly on point-valued observations, the lifted formulation replaces each covariate by a local Gaussian cloud and evaluates the expected hinge loss under this distribution. The resulting objective has an exact closed form, is differentiable for nondegenerate lifting covariances, and approaches the ordinary hinge-loss objective as the covariance scale vanishes.

The lifted formulation leads naturally to clouded representatives that communicate both location

and covariance information. We considered two class-wise constructions. Clouded MR provides a model-independent summary of local class location and variability, whereas clouded SMR additionally matches the local gradient of the lifted objective at its construction parameter. These representatives are exchanged in place of raw observations, gradients, or model parameters and are incorporated directly into the local empirical-risk objectives.

Our theoretical analysis characterizes the approximation induced by frozen clouded SMR representatives. Under boundedness and nondegeneracy conditions, the representative covariance scale remains within first order of the original covariance scale as the within-group covariate diameter decreases. This structural control yields a corresponding first-order bound on the difference between the representative and full-data gradients. Under strong convexity and smoothness, the resulting iterative procedure can be viewed as inexact gradient descent, with an optimization error floor controlled by the representative approximation error. Moreover, score matching makes the first update following a synchronized representative reconstruction exact with respect to the corresponding full-data gradient.

The numerical experiments support these theoretical distinctions. For linear SVMs, the lifted objective preserves classification performance while providing a smooth objective for first-order optimization. Under heterogeneous class allocations, lifted MR and lifted SMR remain competitive with centralized solutions when parameter-aggregation methods are affected by repeated local optimization. Their different constructions become most apparent when communication is infrequent: MR remains stable as a model-independent summary, whereas SMR benefits from periodic reconstruction as the model moves away from its score-matching point. Although MR can achieve predictive performance comparable to SMR, the parameter-recovery experiments show that lifted SMR more closely reproduces the centralized lifted-SVM optimizer.

The same qualitative behavior extends to nonlinear SVMs. A shared kernel dictionary provides a common finite-dimensional coordinate system in which nodes can construct, communicate, and evaluate kernel representatives. Across the Ionosphere and Australian Credit Approval experiments, the representative methods remain competitive with the corresponding centralized shared-dictionary objectives, while RKHS model-distance comparisons again distinguish predictive performance from recovery of the centralized optimizer. Experiments using Random Fourier Features further show that representative learning can also be combined with an alternative finite-dimensional approximation of the RBF kernel.

Taken together, these results demonstrate that lifting and representative construction address complementary aspects of distributed SVM learning. Lifting supplies a differentiable objective that retains the large-margin classification structure, while clouded representatives provide compact distribution-valued summaries through which heterogeneous nodes can exchange local data and objective information. This combination offers a data-centric alternative to repeated parameter aggregation for distributed large-margin learning.

5.2 Future Work

Several directions follow naturally from the present framework. First, the kernel construction currently assumes a common dictionary available to all nodes. In the experiments, this dictionary is formed from a shared subset of the training data. A more fully representative-based construction would build the dictionary itself from communicated summaries rather than from shared observations. For example, class-wise MR or SMR locations could provide initial dictionary elements, while multiple representatives per class could be used when a single summary is insufficient to capture local nonlinear structure. Such a construction would remove the present reliance on shared training observations and would make both the model representation and the distributed optimization depend entirely on representative communication.

Second, the theoretical analysis can be extended beyond the synchronized, locally strongly convex setting considered here. The present gradient discrepancy result treats representatives as frozen within each local optimization phase, and the iterative convergence result assumes a common construction parameter and uniform control of the representative error. An important next step is to analyze node-specific construction parameters, where representatives may be generated from different local iterates, and to quantify explicitly how model disagreement and representative staleness enter the convergence bound. Related questions arise under asynchronous communication, partial participation, and decentralized network topologies.

Third, clouded representatives suggest a broader extension of Representative Learning from point summaries to structured probability-valued summaries. The SVM construction studied here retains local location and covariance information because both quantities enter naturally into the Gaussian-lifted hinge loss. More generally, distributional representatives could encode within-node uncertainty, multimodality, or local geometry that cannot be captured by a single pseudo-observation. Developing principled constructions for richer representative families, together with approximation guarantees that relate their complexity to local heterogeneity, is a natural direction for extending Lifted ERM beyond the SVM setting.

Finally, the distinction between model-independent and gradient-matching representatives is relevant to more expressive learning systems. A representative need not reproduce the gradient of an entire model; it may instead be constructed to preserve task-relevant information at a selected feature representation or layer. This suggests studying representative construction in learned embedding spaces and other structured feature spaces, where compact synthetic summaries could support distributed optimization without requiring direct exchange of the full model state. The present SVM results provide a tractable setting in which this principle can be studied analytically, while extensions to richer model classes remain an important direction for future work.

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